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Gene therapy for ocular disorders

Abstract

Compositions and methods are provided for treating ocular disorders in a subject are provided. In one aspect, an adeno-associated viral vector is provided which includes a nucleic acid molecule comprising a sequence encoding CNGA3. In another aspect, an adeno-associated viral vector is provided which includes a nucleic acid molecule comprising a sequence encoding CNGB3. In another aspect, an adeno-associated viral vector is provided which includes a nucleic acid molecule comprising a sequence encoding REP-1. In desired embodiments, the subject is human, cat, dog, sheep, or non-human primate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This Application is a divisional of U.S. patent application Ser. No. 16/061,530, filed Jun. 12, 2018, which is a US National Stage of PCT/US2016/066402, filed Dec. 13, 2016, and which claims priority to U.S. Provisional Patent Application No. 62/266,789, filed Dec. 14, 2015. Each of these applications are incorporated herein by reference.

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED IN ELECTRONIC FORM (1) Applicant hereby incorporates by reference the Sequence Listing material filed in electronic form herewith. This file is labeled "16-7660_Seq_Listing_new_ST25.txt".

BACKGROUND OF THE INVENTION

(2) Choroideremia (CHM) is an X-linked inherited retinal disease characterized by the degeneration of photoreceptors, retinal pigment epithelium (RPE) and choriocapillaris. Symptoms develop in the 1st or 2nd decade of life with complaints of poor night vision (nyctalopia) and progressive loss of peripheral vision. Visual fields constrict as the disease progresses. This culminates with loss of central vision (visual acuity) and blindness as early as the fourth decade of life. More than 140 mutations in the CHM gene have been found to cause choroideremia. Mutations may lead to the production of an abnormally small, nonfunctional and/or unstable Rab escort protein-1 (REP-1) protein, a decrease in the protein's function or loss of REP-1 protein production. Lack of normal REP-1 disrupts the ability of Rab proteins to aid in intracellular trafficking. The immobility of proteins and organelles within the cell causes the cells to function poorly and to die prematurely.

(3) The choroideremia gene, CHM, encodes Rab Escort Protein-1 (REP-1), a 653 amino acid protein involved in regulation of membrane trafficking. Since the CHM locus is on the X-chromosome, choroideremia is typically only diagnosed in males. Although female carriers of the disease are usually asymptomatic, retinal exams often reveal a patchy degeneration of the retina and RPE and female individuals can be affected depending on the extent of X-inactivation of the normal X chromosome (lyonization). Coussa, RG, Traboulsi, EI (2012) Choroideremia: a review of general findings and pathogenesis, *Ophthalmic Genet* 33(2): 57-65, which is incorporated herein by reference. See also, Vasireddy et al, AAV-mediated gene therapy for choroideremia: preclinical studies in personalized models. *PLoS One*. 2013 May 7;8(5):e61396, which is incorporated herein by reference.

(4) Achromatopsia is a heterogeneous group of autosomal recessive inherited retinal diseases characterized by early onset reduced visual acuity, impaired or complete color blindness, nystagmus, photoaversion and loss of cone photoreceptor function. About 80% of achromatopsia patients show mutations in the alpha or beta subunit (A3 and B3) of the cGMP controlled cation channel cyclic nucleotide-gated channel (CNG) of cone photoreceptors. Homologous to the human disease, Cnga3 deficient mice reveal a loss of cone specific functionality leading to malfunction and degeneration of affected cone photoreceptors.

(5) Therefore, compositions useful for expressing CNGA3 or CNGB3 in human subjects are needed.

SUMMARY OF THE INVENTION

(6) Choroideremia (CHM) is an X-linked retinal degeneration that is symptomatic in the 1st or 2nd decade of life causing nyctalopia and loss of peripheral vision. The disease progresses through mid-life, when most patients become blind. CHM is a favorable target for gene augmentation therapy, as the disease is due to loss of function of a protein necessary for retinal cell health, Rab Escort Protein 1 (REP1), which is encoded by the CHM gene. The CHM cDNA can be packaged in recombinant adeno-associated virus (rAAV), which has an established track record in human gene therapy studies. In addition, there are sensitive and quantitative assays to document REP1 activity, including its ability to prenylate Rab proteins such as Rab27 and to correct a defect in Rab27 localization and trafficking due to lack of prenylation in REP-1 deficient cells.

(7) In one aspect, a codon optimized cDNA sequence encoding Rab Escort Protein-1 (REP-1) is provided. In one embodiment, the codon optimized cDNA sequence is a variant of SEQ ID NO: 3. In another embodiment, the codon optimized cDNA sequence is SEQ ID NO: 1. In another embodiment, the cDNA sequence is codon optimized for expression in humans.

(8) In another aspect, an expression cassette includes a codon optimized nucleic acid sequence that encodes REP-1. In one embodiment, the expression cassette includes the cDNA sequence of SEQ ID NO: 1. In still other embodiments, the REP-1 encoding sequence is positioned between 5' and 3' AAV ITR sequences.

(9) In another embodiment, an adeno-associated virus (AAV) vector is provided. The AAV vector

includes an AAV capsid and a nucleic acid sequence comprising AAV inverted terminal repeat sequences and a nucleic acid sequence encoding human Rab Escort Protein-1 (REP-1), and expression control sequences that direct expression of the REP-1 in a host cell. In one embodiment, the REP-1 sequence encodes a full length REP-1 protein. In one embodiment, the REP-1 sequence is the protein sequence of SEQ ID NO: 2.

(10) In one aspect, a codon optimized cDNA sequence encoding cyclic nucleotide gated channel alpha 3 (CNGA3) is provided. In one embodiment, the codon optimized cDNA sequence is a variant of SEQ ID NO: 13 or SEQ ID NO: 15. In another embodiment, the codon optimized cDNA sequence is SEQ ID NO: 9 or SEQ ID NO: 11. In another embodiment, the cDNA sequence is codon optimized for expression in humans.

(11) In another aspect, an expression cassette includes a codon optimized nucleic acid sequence that encodes cyclic nucleotide gated channel alpha 3 (CNGA3). In one embodiment, the expression cassette includes the cDNA sequence of SEQ ID NO: 9, SEQ ID NO: 11, SEQ ID NO: 13, or SEQ ID NO: 15. In still other embodiments, the CNGA3 encoding sequence is positioned between 5' and 3' AAV ITR sequences.

(12) In another aspect, an expression cassette includes a codon optimized nucleic acid sequence that encodes cyclic nucleotide gated channel alpha 3 (CNGB3). In one embodiment, the expression cassette includes the cDNA sequence of SEQ ID NO: 19 or SEQ ID NO: 21 or SEQ ID NO: 23. In still other embodiments, the CNGB3 encoding sequence is positioned between 5' and 3' AAV ITR sequences.

(13) In another embodiment, an adeno-associated virus (AAV) vector is provided. The AAV vector includes an AAV capsid and a nucleic acid sequence comprising AAV inverted terminal repeat sequences and a nucleic acid sequence encoding human CNGA3, and expression control sequences that direct expression of the CNGA3 in a host cell. In one embodiment, the CNGA3 sequence encodes a full length CNGA3 protein. In one embodiment, the CNGA3 sequence is the protein sequence of SEQ ID NO: 10, SEQ ID NO: 12 or SEQ ID NO: 14.

(14) In another embodiment, an adeno-associated virus (AAV) vector is provided. The AAV vector includes an AAV capsid and a nucleic acid sequence comprising AAV inverted terminal repeat sequences and a nucleic acid sequence encoding human CNGB3, and expression control sequences that direct expression of the CNGB3 in a host cell. In one embodiment, the CNGB3 sequence encodes a full length CNGB3 protein. In one embodiment, the CNGB3 sequence is the protein sequence of SEQ ID NO: 20.

(15) In another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV8 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding REP-1, inverted terminal repeat sequences and expression control sequences that direct expression of REP-1 in a host cell.

(16) In yet another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV8 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding CNGA3, inverted terminal repeat sequences and expression control sequences that direct expression of CNGA3 in a host cell.

(17) In yet another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV8 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding CNGB3, inverted terminal repeat sequences and expression control sequences that direct expression of CNGB3 in a host cell.

(18) In another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV2 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding REP-1, inverted terminal repeat sequences and expression control sequences that direct expression of REP-1 in a host cell.

(19) In yet another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV2 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid

sequences encoding CNGA3, inverted terminal repeat sequences and expression control sequences that direct expression of CNGA3 in a host cell.

(20) In yet another aspect, an adeno-associated virus (AAV) vector is provided which comprises an AAV2 capsid and an expression cassette, wherein said expression cassette comprises nucleic acid sequences encoding CNGB3, inverted terminal repeat sequences and expression control sequences that direct expression of CNGB3 in a host cell.

(21) In another aspect, a pharmaceutical composition is provided which includes a pharmaceutically acceptable carrier, diluent, excipient and/or adjuvant and a least a viral vector as described herein.

(22) In yet a further aspect a pharmaceutical composition comprises a pharmaceutically acceptable carrier, diluent, excipient and/or adjuvant and the nucleic acid sequence, a plasmid, a vector, or a viral vector, such as the rAAV, described specifically herein.

(23) In another aspect, a method for treating choroideremia is provided. In one embodiment, the method includes administering a composition which includes the AAV vector which encodes REP-1, as described herein, to a subject in need thereof.

(24) In another aspect, a method for treating achromatopsia is provided. In one embodiment, the method includes administering a composition which includes the AAV vector which encodes CNGA3, as described herein, to a subject in need thereof.

(25) In another aspect, a method for treating achromatopsia is provided. In one embodiment, the method includes administering a composition which includes the AAV vector which encodes CNGB3, as described herein, to a subject in need thereof.

(26) In yet another aspect, a plasmid for producing an AAV vector is provided. In one embodiment, the plasmid includes the codon optimized cDNA sequence encoding REP-1 as described herein. In another embodiment, the plasmid includes the codon optimized cDNA sequence encoding CNGA3 as described herein. In another embodiment, the plasmid includes a codon optimized cDNA sequence encoding CNGB3 which is a sequence sharing at least 70% identity with SEQ ID NO: 19 or SEQ ID NO: 21. In one embodiment, the plasmid is modular.

(27) In another aspect, a method of generating a rAAV virus is provided. The method includes culturing a packaging cell carrying the plasmid described herein in the presence of sufficient viral sequences to permit packaging of the gene expression cassette viral genome into an infectious AAV envelope or capsid. In another, aspect, a recombinant AAV produced according to the method is provided.

(28) Other aspects and advantages of the invention will be readily apparent from the following detailed description of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A and FIG. 1B are gels showing REP-1 protein expression in vitro after transfection of cultured 84-31 HEK cells. The first lane of each gel shows expression of codon-optimized REP-1 as described herein, expressed from plasmid p944. The second lane shows expression of native REP-1 from plasmid p742. The third lane shows endogenous expression of REP-1 by 84-31 cells that were not transfected with a plasmid. The last lane is a blank. The gels demonstrate that the codon-optimized REP-1 sequence, as described herein, results in a higher level of protein expression than the native REP-1 sequence, and that levels of expression from the exogenously transfected plasmids are many-fold higher than endogenous REP-1 expression.

(2) FIG. 2 is an alignment of the native REP-1 coding sequence of SEQ ID NO: 1 vs. the codon optimized REP-1 coding sequence of SEQ ID NO: 3.

(3) FIG. 3 is an alignment of the native CNGA3 coding sequence of SEQ ID NO: 13 vs. the codon

optimized CNGA3 coding sequence of SEQ ID NO: 9.

(4) FIG. 4 is an alignment of CNGB3 native ORF (SEQ ID NO: 19) vs. CNGB3 modified ORF (SEQ ID NO: 21) vs. CNGB3 modified orf with modified ends (SEQ ID NO: 23). Point mutations are highlighted.

(5) FIG. 5 is a plasmid map of p584, described herein. The sequence of p584 is shown in SEQ ID NO: 7.

(6) FIG. 6 is a plasmid map of AAV.hCHMco. Version 2a, described herein. The sequence of Version 2a is shown in SEQ ID NO: 25.

(7) FIG. 7 is a plasmid map of AAV.hCHMco. Version 2b, described herein. The sequence of Version 2b is shown in SEQ ID NO: 26.

(8) FIG. 8 is a plasmid map of AAV.hCHMco. Version 3a, described herein. The sequence of Version 3a is shown in SEQ ID NO: 27.

(9) FIG. 9 is a plasmid map of AAV.hCHMco. Version 3b, described herein. The sequence of Version 3b is shown in SEQ ID NO: 28.

(10) FIG. 10 is a plasmid map of AAV.hCHM. Version 1, described herein. The sequence of Version 1 is shown in SEQ ID NO: 29. FIG. 11 is a graphic representation of the effect of lambda insert on AAV product impurity. All a-version (lambda containing) vectors have much reduced Kan⁺ signals from qPCR test.

(11) FIG. 12A is a western blot showing human anti-REP-1 antibody detection of a protein of ~75-80 kDa in ocular tissues of CD-1 mice injected with AAV8.2b at 5E9 (High dose) vector genome copies. Animals injected with AAV8.2b at 5E8 (Low dose) showed a very faint protein band at ~75-80 kDa. FIG. 12B is a Western blot analysis of ocular tissues of AAV8.3b injected CD1 mice (2 mice/group) detected with anti-REP-1 antibody, which revealed the presence of a protein of ~75-80 kDa in one eye injected with low dose and in both eyes injected with high dose of AAV8.3b. In the ocular tissues of uninjected mice there was no REP-1 expression detected.

DETAILED DESCRIPTION OF THE INVENTION

(12) The methods and compositions described herein involve compositions and methods for delivering optimized CHM encoding REP-1 to mammalian subjects for the treatment of ocular disorders, primarily blinding diseases such as choroideremia. In addition, methods and compositions described herein involve compositions and methods for delivering optimized CNGA3 or CNGB3 to mammalian subjects for the treatment of ocular disorders, primarily blinding diseases such as achromatopsia. In one embodiment, such compositions involve codon optimization of the REP-1, CNGA3 or CNGB3 coding sequence. It is believed that these features increase the efficacy of the product, and increase safety, since a lower dose of reagent is used. It is anticipated that this optimization of the transgene cassette could theoretically maximize the level of production of the experimental protein compared to levels that can be generated using the endogenous sequence. However, also encompassed herein are compositions which include the native REP1, CNGA3, and CNGB3 coding sequences, as shown in SEQ ID NO: 3, SEQ ID NO: 13 and SEQ ID NO: 19, respectively. It is to be understood that when an embodiment is described for either REP-1, CNGA3 or CNGB3, a similar embodiment is intended to be recited for the other.

(13) Technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs and by reference to published texts, which provide one skilled in the art with a general guide to many of the terms used in the present application. The definitions contained in this specification are provided for clarity in describing the components and compositions herein and are not intended to limit the claimed invention.

(14) The choroideremia gene, CHM, encodes Rab Escort Protein-1 (REP-1), a 653 amino acid protein thought to be involved in membrane trafficking. As used herein, the terms "REP-1" and "CHM" are used interchangeably when referring to the coding sequence. Since the CHM locus is on the X-chromosome, choroideremia is typically only diagnosed in males. Although female carriers of the disease are usually asymptomatic, retinal exams often reveal a patchy degeneration

of the retina and RPE and female individuals can be affected depending on the extent of X-inactivation of the normal X chromosome (lyonization). See, Coussa, cited above. The native amino acid sequence encoding human REP-1 is reported at GenBank accession number P24386, and reproduced here in SEQ ID NO: 2. The native human nucleic acid sequence of CHM is reproduced here at SEQ ID NO: 3 (accession no. NM_000390.2).

(15) Cyclic nucleotide-gated (CNG) ion channels are key mediators underlying signal transduction in retinal and olfactory receptors. Genetic defects in CNGA3 and CNGB3, encoding two structurally related subunits of cone CNG channels, are known to lead to achromatopsia. CNGA3 is a 694 amino acid protein. CNGB is an 809 amino acid protein.

(16) Achromatopsia is a heterogeneous group of congenital, autosomal recessive retinal disorders that manifest by early onset cone photoreceptor dysfunction, severely reduced visual acuity, impaired or complete color blindness and photophobia. The native nucleic acid sequence encoding human CNGA3 is reported at GenBank accession no. XM_011210554.1, and reproduced in SEQ ID NO: 13. The native nucleic acid sequence encoding human CNGB3 is reported at GenBank accession no. XM_011210554.1, and reproduced in SEQ ID NO: 13. The native nucleic acid sequence for the human CNGA3 X1 variant, which includes an additional exon, is reported at GenBank accession no. NM_001298.2, and reproduced in SEQ ID NO: 15. The native nucleic acid sequence encoding human CNGB3 is reproduced in SEQ ID NO: 19.

(17) In certain embodiments of this invention, a subject has an “ocular disorder”, for which the components, compositions and methods of this invention are designed to treat. As used herein, the term “subject” as used herein means a mammalian animal, including a human, a veterinary or farm animal, a domestic animal or pet, and animals normally used for clinical research. In one embodiment, the subject of these methods and compositions is a human. Still other suitable subjects include, without limitation, murine, rat, canine, feline, porcine, bovine, ovine, non-human primate and others. As used herein, the term “subject” is used interchangeably with “patient”.

(18) As used herein “ocular disorder” includes, cone-rod dystrophies and retinal diseases including, without limitation. Stargardt disease (autosomal dominant or autosomal recessive), retinitis pigmentosa, and pattern dystrophy. In one embodiment, the subject has achromatopsia. In another embodiment, the subject has choroideremia or an X-linked hereditary retinal degeneration. Clinical signs of such ocular diseases include, but are not limited to, decreased peripheral vision, decreased central (reading) vision, decreased night vision, loss of color perception, reduction in visual acuity, decreased photoreceptor function, pigmentary changes, and ultimately blindness.

(19) As used herein, the term “treatment” or “treating” is defined encompassing administering to a subject one or more compounds or compositions described herein for the purposes of amelioration of one or more symptoms of an ocular disease. “Treatment” can thus include one or more of reducing onset or progression of an ocular disease, preventing disease, reducing the severity of the disease symptoms, or retarding their progression, including the progression of blindness, removing the disease symptoms, delaying onset of disease or monitoring progression of disease or efficacy of therapy in a given subject.

(20) The term “exogenous” as used to describe a nucleic acid sequence or protein means that the nucleic acid or protein does not naturally occur in the position in which it exists in a chromosome, or host cell. An exogenous nucleic acid sequence also refers to a sequence derived from and inserted into the same host cell or subject, but which is present in a non-natural state, e.g. a different copy number, or under the control of different regulatory elements.

(21) The term “heterologous” as used to describe a nucleic acid sequence or protein means that the nucleic acid or protein was derived from a different organism or a different species of the same organism than the host cell or subject in which it is expressed. The term “heterologous” when used with reference to a protein or a nucleic acid in a plasmid, expression cassette, or vector, indicates that the protein or the nucleic acid is present with another sequence or subsequence with which the protein or nucleic acid in question is not found in the same relationship to each other in

nature.

(22) The terms “percent (%) identity”, “sequence identity”, “percent sequence identity”, or “percent identical” in the context of nucleic acid sequences refers to the bases in the two sequences which are the same when aligned for correspondence. The percent identity is determined by comparing two sequences aligned under optimal conditions over the sequences to be compared. The length of sequence identity comparison may be over the full-length of the REP-1, CNGA3 or CNGB3 coding sequence, or a fragment of at least about 100 to 150 nucleotides, or as desired. However, identity among smaller fragments, e.g. of at least about nine nucleotides, usually at least about 20 to 24 nucleotides, at least about 28 to 32 nucleotides, at least about 36 or more nucleotides, may also be desired. Multiple sequence alignment programs are also available for nucleic acid sequences. Examples of such programs include, “Clustal W”, “CAP Sequence Assembly”, “BLAST”, “MAP”, and “MEME”, which are accessible through Web Servers on the internet. Other sources for such programs are known to those of skill in the art. Alternatively, Vector NTI utilities are also used. There are also a number of algorithms known in the art that can be used to measure nucleotide sequence identity, including those contained in the programs described above. As another example, polynucleotide sequences can be compared using Fasta™, a program in GCG Version 6.1. Commonly available sequence analysis software, more specifically, BLAST or analysis tools provided by public databases may also be used.

(23) The term “isolated” means that the material is removed from its original environment (e.g., the natural environment if it is naturally occurring). For example, a naturally-occurring polynucleotide or polypeptide present in a living animal is not isolated, but the same polynucleotide or polypeptide, separated from some or all of the coexisting materials in the natural system, is isolated, even if subsequently reintroduced into the natural system. Such polynucleotides could be part of a vector and/or such polynucleotides or polypeptides could be part of a composition, and still be isolated in that such vector or composition is not part of its natural environment.

(24) By “engineered” is meant that the nucleic acid sequences encoding the REP-1 or CNGA3 or CNGB3 protein described herein are assembled and placed into any suitable genetic element, e.g., naked DNA, phage, transposon, cosmid, episome, etc., which transfers the REP-1 or CNGA3 or CNGB3 sequences carried thereon to a host cell, e.g., for generating non-viral delivery systems (e.g., RNA-based systems, naked DNA, or the like) or for generating viral vectors in a packaging host cell and/or for delivery to a host cells in a subject. In one embodiment, the genetic element is a plasmid. The methods used to make such engineered constructs are known to those with skill in nucleic acid manipulation and include genetic engineering, recombinant engineering, and synthetic techniques. See, e.g., Green and Sambrook, *Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Press, Cold Spring Harbor, NY (2012).

(25) The term “transgene” as used herein means an exogenous or engineered protein-encoding nucleic acid sequence that is under the control of a promoter or expression control sequence in an expression cassette, rAAV genome, recombinant plasmid or production plasmid, vector, or host cell described in this specification. In certain embodiments, the transgene is a human CHM (REP-1) sequence, encoding a functional REP-1 protein. In some embodiments, the transgene is a codon optimized nucleic acid CHM (REP-1) encoding the REP-1 amino acid sequence set forth in SEQ ID NO: 2. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 1. In certain embodiments, the REP-1 transgene is encoded by the sequence set forth in SEQ ID NO: 5. SEQ ID NO: 5 includes modified ends, which include restriction sites for cloning into a plasmid, such as a production plasmid described herein.

(26) In certain embodiments, the transgene is a human CNGA3 sequence, encoding a functional CNGA3 protein. In certain embodiments, the transgene is a codon optimized CNGA3 encoding sequence SEQ ID NO: 10. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 9. In one embodiment, the transgene includes modified ends, such as that shown in SEQ ID NO: 16, SEQ IDNO 17 or SEQ ID NO: 18, which include restriction sites for

cloning into a plasmid, such as a plasmid described herein. In certain embodiments, the transgene is a codon optimized CNGA3 encoding sequence SEQ ID NO: 12. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 11. In certain embodiments, the transgene is encoded by the native coding sequence of CNGA3, which is set forth in SEQ ID NO: 13.

(27) In certain embodiments, the transgene is a human CNGB3 sequence, encoding a functional CNGB3 protein. In certain embodiments, the transgene is a codon optimized CNGB3 encoding sequence which is a sequence sharing at least 70% identity with SEQ ID NO: 19 or 21. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 23. SEQ ID NO: 23 includes modified ends, which include restriction sites for cloning into a plasmid, such as a production plasmid described herein. Nucleotides 13 to 2448 of SEQ ID NO: 23 provide the ORF for CNGB3. In certain embodiments, the transgene is a codon optimized CNGB3 encoding sequence SEQ ID NO: 20. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 19. In certain embodiments, the transgene is encoded by the sequence set forth in SEQ ID NO: 21. In certain embodiments, the transgene includes modified ends for cloning into a plasmid, such as the plasmids described herein. SEQ ID NO: 21 is a novel cDNA sequence in which certain silent mutations have been made to the native coding sequence. Further modifications to the native sequence, as described herein, are contemplated by the invention.

(28) In one embodiment, the nucleic acid sequence encoding REP-1, CNGA or CNGB further comprises a nucleic acid encoding a tag polypeptide covalently linked thereto. The tag polypeptide may be selected from known “epitope tags” including, without limitation, a myc tag polypeptide, a glutathione-S-transferase tag polypeptide, a green fluorescent protein tag polypeptide, a myc-pyruvate kinase tag polypeptide, a His6 tag polypeptide, an influenza virus hemagglutinin tag polypeptide, a flag tag polypeptide, and a maltose binding protein tag polypeptide.

(29) A “vector” as used herein is a nucleic acid molecule into which an exogenous or heterologous or engineered nucleic acid transgene may be inserted which can then be introduced into an appropriate host cell. Vectors preferably have one or more origin of replication, and one or more site into which the recombinant DNA can be inserted. Vectors often have convenient means by which cells with vectors can be selected from those without, e.g., they encode drug resistance genes. Common vectors include plasmids, viral genomes, and (primarily in yeast and bacteria) “artificial chromosomes.” Certain plasmids are described herein.

(30) “Virus vectors” are defined as replication defective viruses containing the exogenous or heterologous CHM (REP-1) or CNGA3 or CNGB3 nucleic acid transgene(s). In one embodiment, an expression cassette as described herein may be engineered onto a plasmid which is used for drug delivery or for production of a viral vector. Suitable viral vectors are preferably replication defective and selected from amongst those which target ocular cells. Viral vectors may include any virus suitable for gene therapy, including but not limited to adenovirus; herpes virus; lentivirus; retrovirus; parvovirus, etc. However, for ease of understanding, the adeno-associated virus is referenced herein as an exemplary virus vector.

(31) A “replication-defective virus” or “viral vector” refers to a synthetic or recombinant viral particle in which an expression cassette containing a gene of interest is packaged in a viral capsid or envelope, where any viral genomic sequences also packaged within the viral capsid or envelope are replication-deficient; i.e., they cannot generate progeny virions but retain the ability to infect target cells. In one embodiment, the genome of the viral vector does not include genes encoding the enzymes required to replicate (the genome can be engineered to be “gutless”—containing only the transgene of interest flanked by the signals required for amplification and packaging of the artificial genome), but these genes may be supplied during production. Therefore, it is deemed safe for use in gene therapy since replication and infection by progeny virions cannot occur except in the presence of the viral enzyme required for replication.

(32) In still another embodiment, the expression cassette, including any of those described herein is

employed to generate a recombinant AAV genome.

(33) As used herein, the term “host cell” may refer to the packaging cell line in which a recombinant AAV is produced from a production plasmid. In the alternative, the term “host cell” may refer to any target cell in which expression of the transgene is desired. Thus, a “host cell,” refers to a prokaryotic or eukaryotic cell that contains exogenous or heterologous DNA that has been introduced into the cell by any means, e.g., electroporation, calcium phosphate precipitation, microinjection, transformation, viral infection, transfection, liposome delivery, membrane fusion techniques, high velocity DNA-coated pellets, viral infection and protoplast fusion.

(34) In certain embodiments herein, the term “host cell” refers to cultures of ocular cells of various mammalian species for in vitro assessment of the compositions described herein. In other embodiments herein, the term “host cell” refers to the cells employed to generate and package the viral vector or recombinant virus. Still in other embodiments, the term “host cell” is intended to reference the ocular cells of the subject being treated in vivo for the ocular disease.

(35) As used herein, the term “ocular cells” refers to any cell in, or associated with the function of, the eye. The term may refer to any one of photoreceptor cells, including rod photoreceptors, cone photoreceptors and photosensitive ganglion cells, retinal pigment epithelium (RPE) cells, Mueller cells, choroidal cells, bipolar cells, horizontal cells, and amacrine cells. In one embodiment, the ocular cells are the photoreceptor cells. In another embodiment, the ocular cells are RPE cells.

(36) “Plasmids” generally are designated herein by a lower case p preceded and/or followed by capital letters and/or numbers, in accordance with standard naming conventions that are familiar to those of skill in the art. Many plasmids and other cloning and expression vectors that can be used in accordance with the present invention are well known and readily available to those of skill in the art. Moreover, those of skill readily may construct any number of other plasmids suitable for use in the invention. The properties, construction and use of such plasmids, as well as other vectors, in the present invention will be readily apparent to those of skill from the present disclosure.

(37) As used herein, the term “transcriptional control sequence” or “expression control sequence” refers to DNA sequences, such as initiator sequences, enhancer sequences, and promoter sequences, which induce, repress, or otherwise control the transcription of protein encoding nucleic acid sequences to which they are operably linked.

(38) As used herein, the term “operably linked” or “operatively associated” refers to both expression control sequences that are contiguous with the nucleic acid sequence encoding the REP-1 or CNGA3 and/or expression control sequences that act in trans or at a distance to control the transcription and expression thereof.

(39) The term “AAV” or “AAV serotype” as used herein refers to the dozens of naturally occurring and available adeno-associated viruses, as well as artificial AAVs. Among the AAVs isolated or engineered from human or non-human primates (NHP) and well characterized, human AAV2 is the first AAV that was developed as a gene transfer vector; it has been widely used for efficient gene transfer experiments in different target tissues and animal models. Unless otherwise specified, the AAV capsid, ITRs, and other selected AAV components described herein, may be readily selected from among any AAV, including, without limitation, AAV1, AAV2, AAV3, AAV4, AAV5, AAV6, AAV7, AAV8, AAV9, AAV8 bp, AAV7M8 and AAVAnc80, variants of any of the known or mentioned AAVs or AAVs yet to be discovered or variants or mixtures thereof. See, e.g., WO 2005/033321, which is incorporated herein by reference. In another embodiment, the AAV capsid is an AAV8 bp capsid, which preferentially targets bipolar cells. See, WO 2014/024282, which is incorporated herein by reference. In another embodiment, the AAV capsid is an AAV7m8 capsid, which has shown preferential delivery to the outer retina. See, Dalkara et al, In Vivo-Directed Evolution of a New Adeno-Associated Virus for Therapeutic Outer Retinal Gene Delivery from the Vitreous, *Sci Transl Med* 5, 189ra76(2013), which is incorporated herein by reference. In one embodiment, the AAV capsid is an AAV8 capsid. In another embodiment, the AAV capsid an AAV9 capsid. In another embodiment, the AAV capsid an AAV5 capsid. In another embodiment, the AAV

capsid an AAV2 capsid.

(40) As used herein, relating to AAV, the term variant means any AAV sequence which is derived from a known AAV sequence, including those sharing at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, at least 99% or greater sequence identity over the amino acid or nucleic acid sequence. In another embodiment, the AAV capsid includes variants which may include up to about 10% variation from any described or known AAV capsid sequence. That is, the AAV capsid shares about 90% identity to about 99.9% identity, about 95% to about 99% identity or about 97% to about 98% identity to an AAV capsid provided herein and/or known in the art. In one embodiment, the AAV capsid shares at least 95% identity with an AAV capsid. When determining the percent identity of an AAV capsid, the comparison may be made over any of the variable proteins (e.g., vp1, vp2, or vp3). In one embodiment, the AAV capsid shares at least 95% identity with the AAV8 vp3. In another embodiment, a self-complementary AAV is used.

(41) The ITRs or other AAV components may be readily isolated or engineered using techniques available to those of skill in the art from an AAV. Such AAV may be isolated, engineered, or obtained from academic, commercial, or public sources (e.g., the American Type Culture Collection, Manassas, VA). Alternatively, the AAV sequences may be engineered through synthetic or other suitable means by reference to published sequences such as are available in the literature or in databases such as, e.g., GenBank, PubMed, or the like. AAV viruses may be engineered by conventional molecular biology techniques, making it possible to optimize these particles for cell specific delivery of nucleic acid sequences, for minimizing immunogenicity, for tuning stability and particle lifetime, for efficient degradation, for accurate delivery to the nucleus, etc.

(42) As used herein, “artificial AAV” means, without limitation, an AAV with a non-naturally occurring capsid protein. Such an artificial capsid may be generated by any suitable technique, using a selected AAV sequence (e.g., a fragment of a vp1 capsid protein) in combination with heterologous sequences which may be obtained from a different selected AAV, non-contiguous portions of the same AAV, from a non-AAV viral source, or from a non-viral source. An artificial AAV may be, without limitation, a pseudotyped AAV, a chimeric AAV capsid, a recombinant AAV capsid, or a “humanized” AAV capsid. Pseudotyped vectors, wherein the capsid of one AAV is replaced with a heterologous capsid protein, are useful in the invention. In one embodiment, AAV2/5 and AAV2/8 are exemplary pseudotyped vectors.

(43) “Self-complementary AAV” refers a plasmid or vector having an expression cassette in which a coding region carried by a recombinant AAV nucleic acid sequence has been designed to form an intra-molecular double-stranded DNA template. Upon infection, rather than waiting for cell mediated synthesis of the second strand, the two complementary halves of scAAV will associate to form one double stranded DNA (dsDNA) unit that is ready for immediate replication and transcription. See, e.g., D M McCarty et al, “Self-complementary recombinant adeno-associated virus (scAAV) vectors promote efficient transduction independently of DNA synthesis”, Gene Therapy, (August 2001), Vol 8, Number 16, Pages 1248-1254. Self-complementary AAVs are described in, e.g., U.S. Pat. Nos. 6,596,535; 7,125,717; and 7,456,683, each of which is incorporated herein by reference in its entirety.

(44) By “administering” as used in the methods means delivering the composition to the target selected cell which is characterized by the ocular disease. In one embodiment, the method involves delivering the composition by subretinal injection to the RPE, photoreceptor cells or other ocular cells. In another embodiment, intravitreal injection to ocular cells is employed. In still another method, injection via the palpebral vein to ocular cells may be employed. Still other methods of administration may be selected by one of skill in the art given this disclosure. By “administering” or “route of administration” is delivery of composition described herein, with or without a pharmaceutical carrier or excipient, of the subject. Routes of administration may be combined, if desired. In some embodiments, the administration is repeated periodically. The pharmaceutical compositions described herein are designed for delivery to subjects in need thereof by any suitable

route or a combination of different routes. Direct delivery to the eye (optionally via ocular delivery, subretinal injection, intra-retinal injection, intravitreal, topical), or delivery via systemic routes, e.g., intraarterial, intraocular, intravenous, intramuscular, subcutaneous, intradermal, and other parental routes of administration. The nucleic acid molecules and/or vectors described herein may be delivered in a single composition or multiple compositions. Optionally, two or more different AAV may be delivered, or multiple viruses [see, e.g., WO20 2011/126808 and WO 2013/049493]. In another embodiment, multiple viruses may contain different replication-defective viruses (e.g., AAV and adenovirus), alone or in combination with proteins.

(45) Certain compositions described herein are isolated, or synthetically or recombinantly engineered nucleic acid sequences that provide novel codon-optimized sequences encoding REP-1 or CNGA3 or CNGB3. In one embodiment, an isolated or engineered codon optimized nucleic acid sequence encoding human REP-1 is provided. In one embodiment, the codon-optimized sequence is SEQ ID NO: 1. In another embodiment, the codon optimized sequence includes N-terminal and C-terminal restriction sites for cloning. In one embodiment, such as that disclosed in SEQ ID NO: 5, the REP-1 coding sequence includes an N-terminal NotI restriction site and a C-terminal BamHI restriction site, in addition to a Kozak consensus sequence. In addition, the codon optimized sequence, in some embodiments, includes one or more additional restriction sites to allow for addition of markers, such as an epitope tag. When aligned with the native nucleic acid sequence, the codon optimized REP-1 may have a percent identity of at least 50%, or at least 60%, or at least 70%, or at least 80% or at least 90%, including any integer between any of those ranges. In one embodiment, the codon optimized REP-1 has a percent identity with the native sequence of at least 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98 or 99%. In one embodiment, when aligned with the native nucleic acid sequence SEQ ID NO: 3, it is revealed that codon optimized REP-1 (SEQ ID NO: 1) has a percent sequence identity of only 74% (see FIG. 2).

(46) In another embodiment, an isolated or engineered codon optimized nucleic acid sequence encoding human CNGA3 is provided. In one embodiment, the codon-optimized sequence is SEQ ID NO: 9. In one embodiment, the codon-optimized sequence is a CNGA3 variant shown in SEQ ID NO: 11. In another embodiment, the codon optimized sequence includes N-terminal and C-terminal restriction sites for cloning. In one embodiment, the CNGA3 coding sequence includes an N-terminal NotI restriction site and a C-terminal BglII restriction site, in addition to a Kozak consensus sequence. Examples of CNGA3 sequences which include such modifications can be found in SEQ ID NO: 16, SEQ ID NO: 17 and SEQ ID NO: 18. In addition, the codon optimized sequence, in some embodiments, includes one or more additional restriction sites to allow for addition of markers, such as an epitope tag. When aligned with the native nucleic acid sequence, the codon optimized CNGA3 may have a percent identity of at least 50%, or at least 60%, or at least 70%, or at least 80% or at least 90%, including any integer between any of those ranges. In one embodiment, the codon optimized CNGA3 has a percent identity with the native sequence of at least 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98 or 99%. In one embodiment, when aligned with the native nucleic acid sequence SEQ ID NO: 13, it is revealed that codon optimized CNGA3 (SEQ ID NO: 9) has a percent sequence identity of only 80% (see FIG. 3).

(47) In another embodiment, an isolated or engineered codon optimized nucleic acid sequence encoding human CNGB3 is provided. In one embodiment, the codon-optimized sequence is a sequence sharing at least 70% identity with SEQ ID NO: 19 or SEQ ID NO 21. In another embodiment, the codon optimized sequence includes N-terminal and C-terminal restriction sites for cloning, for example, as shown in SEQ ID NO: 23. In addition, the codon optimized sequence, in some embodiments, includes one or more additional restriction sites to allow for addition of markers, such as an epitope tag. When aligned with the native nucleic acid sequence (as shown in

SEQ ID NO: 19), the codon optimized CNGB3 may have a percent identity of at least 50%, or at least 60%, or at least 70%, or at least 80% or at least 90%, including any integer between any of those ranges. In one embodiment, the codon optimized CNGB3 has a percent identity with the native sequence of at least 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98 or 99%.

(48) In one embodiment, the optimized nucleic acid sequences encoding the REP-1 or CNGA3 constructs described herein are engineered into any suitable genetic element, e.g., naked DNA, phage, transposon, cosmid, RNA molecule (e.g., mRNA), episome, etc., which transfers the REP-1 or CNGA3 sequences carried thereon to a host cell, e.g., for generating nanoparticles carrying DNA or RNA, viral vectors in a packaging host cell and/or for delivery to a host cells in subject. In one embodiment, the genetic element is a plasmid.

(49) The selected genetic element may be delivered by any suitable method, including transfection, electroporation, liposome delivery, membrane fusion techniques, high velocity DNA-coated pellets, viral infection and protoplast fusion. The methods used to make such constructs are known to those with skill in nucleic acid manipulation and include genetic engineering, recombinant engineering, and synthetic techniques. See, e.g., Green and Sambrook, *Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Press, Cold Spring Harbor, NY (2012).

(50) A variety of expression cassettes are provided which employ SEQ ID NOs. 1 or 5 for expression of the REP-1 protein. In one embodiment, an example of a plasmid containing such an expression cassette is shown in SEQ ID NO. 25. In one embodiment, an example of a plasmid containing such an expression cassette is shown in SEQ ID NO. 26. In one embodiment, an example of a plasmid containing such an expression cassette is shown in SEQ ID NO. 27. In one embodiment, an example of a plasmid containing such an expression cassette is shown in SEQ ID NO. 28. In another embodiment, a variety of expression cassettes are provided which employ SEQ ID NOs. 9, 11 or 13 for expression of the CNGA3 protein. In another embodiment, a variety of expression cassettes are provided which employ SEQ ID NOs. 19, 21 or 23 for expression of the CNGAB protein. As used herein, an "expression cassette" refers to a nucleic acid molecule which comprises coding sequences for the optimized REP-1 or CNGA3 or CNGB3 proteins, promoter, and may include other regulatory sequences therefor, which cassette may be engineered into a genetic element or plasmid, and/or packaged into the capsid of a viral vector (e.g., a viral particle). In one embodiment, an expression cassette comprises a codon optimized nucleic acid sequence that encodes REP-1. In one embodiment, the cassette provides the codon optimized REP-1 operatively associated with expression control sequences that direct expression of the codon optimized nucleic acid sequence that encodes REP-1 in a host cell.

(51) In another embodiment, an expression cassette comprises a codon optimized nucleic acid sequence that encodes CNGA3. In one embodiment, the cassette provides the codon optimized CNGA3 operatively associated with expression control sequences that direct expression of the codon optimized nucleic acid sequence that encodes CNGA3 in a host cell.

(52) In another embodiment, an expression cassette comprises a codon optimized nucleic acid sequence that encodes CNGB3. In one embodiment, the cassette provides the codon optimized CNGB3 operatively associated with expression control sequences that direct expression of the codon optimized nucleic acid sequence that encodes CNGB3 in a host cell.

(53) In another embodiment, an expression cassette for use in an AAV vector is provided. In that embodiment, the AAV expression cassette includes at least one AAV inverted terminal repeat (ITR) sequence. In another embodiment, the expression cassette comprises 5' ITR sequences and 3' ITR sequences. In one embodiment, the 5' and 3' ITRs flank the codon optimized nucleic acid sequence that encodes REP-1 or CNGA3 or CNGB3, optionally with additional sequences which direct expression of the codon optimized nucleic acid sequence that encodes REP-1 or CNGA3 or CNGB3 in a host cell. Thus, as described herein, a AAV expression cassette is meant to describe an

expression cassette as described above flanked on its 5' end by a 5' AAV inverted terminal repeat sequence (ITR) and on its 3' end by a 3' AAV ITR. Thus, this rAAV genome contains the minimal sequences required to package the expression cassette into an AAV viral particle, i.e., the AAV 5' and 3' ITRs. The AAV ITRs may be obtained from the ITR sequences of any AAV, such as described herein. These ITRs may be of the same AAV origin as the capsid employed in the resulting recombinant AAV, or of a different AAV origin (to produce an AAV pseudotype). In one embodiment, the ITR sequences from AAV2, or the deleted version thereof (Δ ITR), are used for convenience and to accelerate regulatory approval. However, ITRs from other AAV sources may be selected. Each rAAV genome can be then introduced into a production plasmid. In one embodiment, the production plasmid is that described herein, or as described in WO2012/158757, which is incorporated herein by reference. Various plasmids are known in the art for use in producing rAAV vectors, and are useful herein. The production plasmids are cultured in the host cells which express the AAV cap and/or rep proteins. In the host cells, each rAAV genome is rescued and packaged into the capsid protein or envelope protein to form an infectious viral particle.

(54) One type of production plasmid is that shown in SEQ ID NO: 7, which is termed p584. This plasmid is used in the examples for generation of the rAAV-REP-1 vector. Such a plasmid is one that contains a 5' AAV ITR sequence; a selected promoter; a polyA sequence; and a 3' ITR; additionally, it also contains a stuffer sequence, such as lambda. In one embodiment, a non-coding lambda stuffer region is included in the vector backbone. The nucleic acid sequence encoding REP-1, CNGA3 or CNGB2 are inserted in place of between the selected promoter and the polyA sequence, or a similar, plasmid. An example of p584 which includes the REP-1 encoding sequence can be found in SEQ ID NO: 8. In another embodiment, the production plasmid is modified to optimized vector plasmid production efficiency. Such modifications include addition of other neutral sequences, or deletion of portion(s) of or the entire lambda stuffer sequence to modulate the level of supercoil of the vector plasmid. Such modifications are contemplated herein. In other embodiments, terminator and other sequences are included in the plasmid.

(55) In still a further embodiment, a recombinant adeno-associated virus (AAV) vector is provided for delivery of the REP-1, CNGA3 and CNGB3 constructs and optimized sequences described herein. An adeno-associated virus (AAV) viral vector is an AAV DNase-resistant particle having an AAV protein capsid into which is packaged nucleic acid sequences for delivery to target cells. An AAV capsid is composed of 60 capsid (cap) protein subunits, VP1, VP2, and VP3, that are arranged in an icosahedral symmetry in a ratio of approximately 1:1:10 to 1:1:20, depending upon the selected AAV. AAVs may be selected as sources for capsids of AAV viral vectors as identified above. See, e.g., US Published Patent Application No. 2007-0036760-A1; US Published Patent Application No. 2009-0197338-A1; EP 1310571. See also, WO 2003/042397 (AAV7 and other simian AAV), U.S. Pat. Nos. 7,790,449 and 7,282,199 (AAV8), WO 2005/033321 and U.S. Pat. No. 7,906,111 (AAV9), and WO 2006/110689, and WO 2003/042397 (rh.10). These documents also describe other AAV which may be selected for generating AAV and are incorporated by reference. In some embodiments, an AAV cap for use in the viral vector can be generated by mutagenesis (i.e., by insertions, deletions, or substitutions) of one of the aforementioned AAV capsids or its encoding nucleic acid. In some embodiments, the AAV capsid is chimeric, comprising domains from two or three or four or more of the aforementioned AAV capsid proteins. In some embodiments, the AAV capsid is a mosaic of Vp1, Vp2, and Vp3 monomers from two or three different AAVs or recombinant AAVs. In some embodiments, an rAAV composition comprises more than one of the aforementioned Caps.

(56) In another embodiment, the AAV capsid includes variants which may include up to about 10% variation from any described or known AAV capsid sequence. That is, the AAV capsid shares about 90% identity to about 99.9% identity, about 95% to about 99% identity or about 97% to about 98% identity to an AAV capsid provided herein and/or known in the art. In one embodiment, the AAV

capsid shares at least 95% identity with an AAV capsid. When determining the percent identity of an AAV capsid, the comparison may be made over any of the variable proteins (e.g., vp1, vp2, or vp3). In one embodiment, the AAV capsid shares at least 95% identity with the AAV8 vp3. In another embodiment, a self-complementary AAV is used. In one embodiment, it is desirable to utilize an AAV capsid, which shows tropism for the desired target cell, e.g., photoreceptors, RPE or other ocular cells. In one embodiment, the AAV capsid is a tyrosine capsid-mutant in which certain surface exposed tyrosine residues are substituted with phenylalanine (F). Such AAV variants are described, e.g., in Mowat et al, Tyrosine capsid-mutant AAV vectors for gene delivery to the canine retina from a subretinal or intravitreal approach, *Gene Therapy* 21, 96-105 (January 2014), which is incorporated herein by reference.

(57) For packaging an expression cassette or rAAV genome or production plasmid into virions, the ITRs are the only AAV components required in cis in the same construct as the transgene. In one embodiment, the coding sequences for the replication (rep) and/or capsid (cap) are removed from the AAV genome and supplied in trans or by a packaging cell line in order to generate the AAV vector. For example, as described above, a pseudotyped AAV may contain ITRs from a source which differs from the source of the AAV capsid. Additionally or alternatively, a chimeric AAV capsid may be utilized. Still other AAV components may be selected. Sources of such AAV sequences are described herein and may also be isolated or engineered obtained from academic, commercial, or public sources (e.g., the American Type Culture Collection, Manassas, VA). Alternatively, the AAV sequences may be obtained through synthetic or other suitable means by reference to published sequences such as are available in the literature or in databases such as, e.g., GenBank®, PubMed®, or the like.

(58) Methods for generating and isolating AAV viral vectors suitable for delivery to a subject are known in the art. See, e.g., U.S. Pat. Nos. 7,790,449; 7,282,199; WO 2003/042397; WO 2005/033321, WO 2006/110689; and U.S. Pat. No. 7,588,772 B2]. In a one system, a producer cell line is transiently transfected with a construct that encodes the transgene flanked by ITRs and a construct(s) that encodes rep and cap. In a second system, a packaging cell line that stably supplies rep and cap is transiently transfected with a construct encoding the transgene flanked by ITRs. In each of these systems, AAV virions are produced in response to infection with helper adenovirus or herpesvirus, requiring the separation of the rAAVs from contaminating virus. More recently, systems have been developed that do not require infection with helper virus to recover the AAV—the required helper functions (i.e., adenovirus E1, E2a, VA, and E4 or herpesvirus UL5, UL8, UL52, and UL29, and herpesvirus polymerase) are also supplied, in trans, by the system. In these newer systems, the helper functions can be supplied by transient transfection of the cells with constructs that encode the required helper functions, or the cells can be engineered to stably contain genes encoding the helper functions, the expression of which can be controlled at the transcriptional or posttranscriptional level.

(59) In yet another system, the transgene flanked by ITRs and rep/cap genes are introduced into insect cells by infection with baculovirus-based vectors. For reviews on these production systems, see generally, e.g., Zhang et al., 2009, “Adenovirus-Adeno-associated virus hybrid for large-scale recombinant adeno-associated virus production,” *Human Gene Therapy* 20:922-929, the contents of which is incorporated herein by reference in its entirety. Methods of making and using these and other AAV production systems are also described in the following U.S. patents, the contents of each of which is incorporated herein by reference in its entirety: U.S. Pat. Nos. 5,139,941; 5,741,683; 6,057,152; 6,204,059; 6,268,213; 6,491,907; 6,660,514; 6,951,753; 7,094,604; 7,172,893; 7,201,898; 7,229,823; and 7,439,065. See generally, e.g., Grieger & Samulski, 2005, “Adeno-associated virus as a gene therapy vector: Vector development, production and clinical applications,” *Adv. Biochem. Engin/Biotechnol.* 99:119-145; Buning et al., 2008, “Recent developments in adeno-associated virus vector technology,” *J. Gene Med.* 10:717-733; and the references cited below, each of which is incorporated herein by reference in its entirety.

(60) The methods used to construct any embodiment of this invention are known to those with skill in nucleic acid manipulation and include genetic engineering, recombinant engineering, and synthetic techniques. See, e.g., Green and Sambrook et al, *Molecular Cloning: A Laboratory Manual*, Cold Spring Harbor Press, Cold Spring Harbor, NY (2012). Similarly, methods of generating rAAV virions are well known and the selection of a suitable method is not a limitation on the present invention. See, e.g., K. Fisher et al, (1993) *J. Virol.*, 70:520-532 and U.S. Pat. No. 5,478,745.

(61) The rAAV vectors comprise an AAV capsid and an AAV expression cassette comprising sequences encoding REP-1 or CNGA3 or CNGB3, such as described above. In certain embodiments, the rAAV expression cassette comprises AAV inverted terminal repeat sequences and a codon optimized nucleic acid sequence that encodes REP-1 or CNGA3 or CNGB3, and expression control sequences that direct expression of the encoded proteins in a host cell. The rAAV expression cassette, in other embodiments, further comprises one or more of an intron, a Kozak sequence, a polyA, and post-transcriptional regulatory elements. Such rAAV vectors for use in pharmaceutical compositions for delivery to the eye, may employ a capsid from any of the many known AAVs identified above.

(62) Other conventional components of the expression cassettes and vectors include other components that can be optimized for a specific species using techniques known in the art including, e.g, codon optimization, as described herein. The components of the cassettes, vectors, plasmids and viruses or other compositions described herein include a promoter sequence as part of the expression control sequences. In another embodiment, the promoter is cell-specific. The term “cell-specific” means that the particular promoter selected for the recombinant vector can direct expression of the optimized REP-1 or CNGA3 or CNGB3 transgene in a particular ocular cell type. In one embodiment, the promoter is specific for expression of the transgene in photoreceptor cells. In another embodiment, the promoter is specific for expression in the rods and cones. In another embodiment, the promoter is specific for expression in the rods. In another embodiment, the promoter is specific for expression in the cones. In one embodiment, the photoreceptor-specific promoter is a human rhodopsin kinase promoter. The rhodopsin kinase promoter has been shown to be active in both rods and cones. See, e.g., Sun et al, *Gene Therapy with a Promoter Targeting Both Rods and Cones Rescues Retinal Degeneration Caused by AIPL1 Mutations*, *Gene Ther.* 2010 January; 17(1): 117-131, which is incorporated herein by reference in its entirety. In one embodiment, the promoter is modified to add one or more restriction sites to facilitate cloning.

(63) In another embodiment, the promoter is a human rhodopsin promoter. In one embodiment, the promoter is modified to include restriction on the ends for cloning. See, e.g, Nathans and Hogness, *Isolation and nucleotide sequence of the gene encoding human rhodopsin*, *PNAS*, 81:4851-5 (August 1984), which is incorporated herein by reference in its entirety. In another embodiment, the promoter is a portion or fragment of the human rhodopsin promoter. In another embodiment, the promoter is a variant of the human rhodopsin promoter.

(64) Other exemplary promoters include the human G-protein-coupled receptor protein kinase 1 (GRK1) promoter (Genbank Accession number AY327580). In another embodiment, the promoter is a 292 nt fragment (positions 1793-2087) of the GRK1 promoter (See, Beltran et al, *Gene Therapy* 2010 17:1162-74, which is hereby incorporated by reference in its entirety). In another preferred embodiment, the promoter is the human interphotoreceptor retinoid-binding protein proximal (IRBP) promoter. In one embodiment, the promoter is a 235 nt fragment of the hIRBP promoter. In one embodiment, the promoter is the RPGR proximal promoter (Shu et al, *IOVS*, May 2102, which is incorporated by reference in its entirety). Other promoters useful in the invention include, without limitation, the rod opsin promoter, the red-green opsin promoter, the blue opsin promoter, the cGMP- β -phosphodiesterase promoter (Qgueta et al, *IOVS*, *Invest Ophthalmol Vis Sci.* 2000 December;41(13): 4059-63), the mouse opsin promoter (Beltran et al 2010 cited above), the rhodopsin promoter (Mussolino et al, *Gene Ther*, July 2011, 18(7): 637-45); the alpha-subunit

of cone transducin (Morrissey et al, BMC Dev, January 2011, 11:3); beta phosphodiesterase (PDE) promoter; the retinitis pigmentosa (RP1) promoter (Nicord et al, J. Gene Med, December 2007, 9(12): 1015-23); the NXNL2/NXNL1 promoter (Lambard et al, PLoS One, October 2010, 5(10):e13025), the RPE65 promoter; the retinal degeneration slow/peripherin 2 (Rds/perph2) promoter (Cai et al, Exp Eye Res. 2010 August;91(2): 186-94); and the VMD2 promoter (Kachi et al, Human Gene Therapy, 2009(20:31-9)). Each of these documents is incorporated by reference herein in its entirety. In another embodiment, the promoter is selected from human human EF1 α promoter, rhodopsin promoter, rhodopsin kinase, interphotoreceptor binding protein (IRBP), cone opsin promoters (red-green, blue), cone opsin upstream sequences containing the red-green cone locus control region, cone transducing, and transcription factor promoters (neural retina leucine zipper (Nrl) and photoreceptor-specific nuclear receptor Nr2e3, bZIP).

(65) In another embodiment, the promoter is a ubiquitous or constitutive promoter. An example of a suitable promoter is a hybrid chicken β -actin (CBA) promoter with cytomegalovirus (CMV) enhancer elements. In another embodiment, the promoter is the CB7 promoter. Other suitable promoters include the human β -actin promoter, the human elongation factor-1 α promoter, the cytomegalovirus (CMV) promoter, the simian virus 40 promoter, and the herpes simplex virus thymidine kinase promoter. See, e.g., Damdindorj et al, (August 2014) A Comparative Analysis of Constitutive Promoters Located in Adeno-Associated Viral Vectors. PLoS ONE 9(8):e106472. Still other suitable promoters include viral promoters, constitutive promoters, regulatable promoters [see, e.g., WO 2011/126808 and WO 2013/04943]. Alternatively a promoter responsive to physiologic cues may be utilized in the expression cassette, rAAV genomes, vectors, plasmids and viruses described herein. In one embodiment, the promoter is of a small size, under 1000 bp, due to the size limitations of the AAV vector. In another embodiment, the promoter is under 400 bp. Other promoters may be selected by one of skill in the art. In one embodiment, the REP-1 construct incorporates a ubiquitous promoter. In another embodiment, the CNGA3 construct incorporates a photoreceptor-specific promoter. In one embodiment, the REP-1 construct includes a CBA promoter with CMV enhancer elements.

(66) In another embodiment, the promoter is an inducible promoter. The inducible promoter may be selected from known promoters including the rapamycin/rapalog promoter, the ecdysone promoter, the estrogen-responsive promoter, and the tetracycline-responsive promoter, or heterodimeric repressor switch. See, Sochor et al, An Autogenously Regulated Expression System for Gene Therapeutic Ocular Applications. Scientific Reports, 2015 Nov. 24; 5:17105 and Daber R, Lewis M., A novel molecular switch. J Mol Biol. 2009 Aug. 28;391(4): 661-70, Epub 2009 Jun. 21 which are both incorporated herein by reference in their entirety.

(67) In other embodiments, the cassette, vector, plasmid and virus constructs described herein contain other appropriate transcription initiation, termination, enhancer sequences, efficient RNA processing signals such as splicing and polyadenylation (polyA) signals; TATA sequences; sequences that stabilize cytoplasmic mRNA; sequences that enhance translation efficiency (i.e., Kozak consensus sequence); introns; sequences that enhance protein stability; and when desired, sequences that enhance secretion of the encoded product. The expression cassette or vector may contain none, one or more of any of the elements described herein. Examples of suitable polyA sequences include, e.g., SV40, bovine growth hormone (bGH), and TK polyA. Examples of suitable enhancers include, e.g., the CMV enhancer, the RSV enhancer, the alpha fetoprotein enhancer, the TTR minimal promoter/enhancer, LSP (TH-binding globulin promoter/alpha1-microglobulin/bikunin enhancer), amongst others. In one embodiment, a Kozak sequence is included upstream of the transgene coding sequence to enhance translation from the correct initiation codon. In another embodiment, CBA exon 1 and intron are included in the expression cassette. In one embodiment, the transgene is placed under the control of a hybrid chicken β actin (CBA) promoter. This promoter consists of the cytomegalovirus (CMV) immediate early enhancer, the proximal chicken β actin promoter, and CBA exon 1 flanked by intron 1 sequences.

(68) In one embodiment, the expression cassette contains a 5' ITR, CBA promoter, CMV enhancer, CBA exon 1 and intron, kozak sequence, human codon optimized CHM sequence (SEQ ID NO: 1), bGH poly A and 3' ITR.

(69) In yet other aspects, these nucleic acid sequences, vectors, expression cassettes and rAAV viral vectors are useful in a pharmaceutical composition, which also comprises a pharmaceutically acceptable carrier, buffer, diluent and/or adjuvant, etc. Such pharmaceutical compositions are used to express the optimized REP-1 or CNGA3 or CNGB3 in the ocular cells through delivery by such recombinantly engineered AAVs or artificial AAVs.

(70) To prepare these pharmaceutical compositions containing the nucleic acid sequences, vectors, expression cassettes and rAAV viral vectors, the sequences or vectors or viral vector is preferably assessed for contamination by conventional methods and then formulated into a pharmaceutical composition suitable for administration to the eye. Such formulation involves the use of a pharmaceutically and/or physiologically acceptable vehicle or carrier, particularly one suitable for administration to the eye, such as buffered saline or other buffers, e.g., HEPES, to maintain pH at appropriate physiological levels, and, optionally, other medicinal agents, pharmaceutical agents, stabilizing agents, buffers, carriers, adjuvants, diluents, etc. For injection, the carrier will typically be a liquid. Exemplary physiologically acceptable carriers include sterile, pyrogen-free water and sterile, pyrogen-free, phosphate buffered saline. A variety of such known carriers are provided in U.S. Pat. No. 7,629,322, incorporated herein by reference. In one embodiment, the carrier is an isotonic sodium chloride solution. In another embodiment, the carrier is balanced salt solution. In one embodiment, the carrier includes tween. If the virus is to be stored long-term, it may be frozen in the presence of glycerol or Tween20.

(71) In one exemplary specific embodiment, the composition of the carrier or excipient contains 180 mM NaCl, 10 mM NaPi, pH7.3 with 0.0001%-0.01% Pluronic F68 (PF68). The exact composition of the saline component of the buffer ranges from 160 mM to 180 mM NaCl. Optionally, a different pH buffer (potentially HEPES, sodium bicarbonate, TRIS) is used in place of the buffer specifically described. Still alternatively, a buffer containing 0.9% NaCl is useful.

(72) Optionally, the compositions of the invention may contain, in addition to the rAAV and/or variants and carrier(s), other conventional pharmaceutical ingredients, such as preservatives, or chemical stabilizers. Suitable exemplary preservatives include chlorobutanol, potassium sorbate, sorbic acid, sulfur dioxide, propyl gallate, the parabens, ethyl vanillin, glycerin, phenol, and parachlorophenol. Suitable chemical stabilizers include gelatin and albumin.

(73) The pharmaceutical compositions containing at least one replication-defective rAAV virus, as described herein, can be formulated with a physiologically acceptable carrier, diluent, excipient and/or adjuvant, for use in gene transfer and gene therapy applications. In the case of AAV viral vectors, quantification of the genome copies ("GC"), vector genomes ("VG"), or virus particles may be used as the measure of the dose contained in the formulation or suspension. Any method known in the art can be used to determine the genome copy (GC) number of the replication-defective virus compositions of the invention. One method for performing AAV GC number titration is as follows: Purified AAV vector samples are first treated with DNase to eliminate un-encapsidated AAV genome DNA or contaminating plasmid DNA from the production process. The DNase resistant particles are then subjected to heat treatment to release the genome from the capsid. The released genomes are then quantitated by real-time PCR using primer/probe sets targeting specific region of the viral genome (usually poly A signal). In another method the effective dose of a recombinant adeno-associated virus carrying a nucleic acid sequence encoding the optimized REP-1 or CNGA3 transgene is measured as described in S.K. McLaughlin et al, 1988 J. Virol., 62:1963, which is incorporated by reference in its entirety.

(74) As used herein, the term "dosage" can refer to the total dosage delivered to the subject in the course of treatment, or the amount delivered in a single unit (or multiple unit or split dosage) administration. The pharmaceutical virus compositions can be formulated in dosage units to

contain an amount of replication-defective virus carrying the codon optimized nucleic acid sequences encoding REP-1 or CNGA3 or CNGB3 as described herein that is in the range of about 1.0×10^9 GC to about 1.0×10^{15} GC including all integers or fractional amounts within the range. In one embodiment, the compositions are formulated to contain at least 1×10^9 , 2×10^9 , 3×10^9 , 4×10^9 , 5×10^9 , 6×10^9 , 7×10^9 , 8×10^9 , or 9×10^9 GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{10} , 2×10^{10} , 3×10^{10} , 4×10^{10} , 5×10^{10} , 6×10^{10} , 7×10^{10} , 8×10^{10} , or 9×10^{10} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{11} , 2×10^{11} , 3×10^{11} , 4×10^{11} , 5×10^{11} , 6×10^{11} , 7×10^{11} , 8×10^{11} , or 9×10^{11} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{12} , 2×10^{12} , 3×10^{12} , 4×10^{12} , 5×10^{12} , 6×10^{12} , 7×10^{12} , 8×10^{12} , or 9×10^{12} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{13} , 2×10^{13} , 3×10^{13} , 4×10^{13} , 5×10^{13} , 6×10^{13} , 7×10^{13} , 8×10^{13} , or 9×10^{13} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{14} , 2×10^{14} , 3×10^{14} , 4×10^{14} , 5×10^{14} , 6×10^{14} , 7×10^{14} , 8×10^{14} , or 9×10^{14} GC per dose including all integers or fractional amounts within the range. In another embodiment, the compositions are formulated to contain at least 1×10^{15} , 2×10^{15} , 3×10^{15} , 4×10^{15} , 5×10^{15} , 6×10^{15} , 7×10^{15} , 8×10^{15} , or 9×10^{15} GC per dose including all integers or fractional amounts within the range. In one embodiment, for human application the dose can range from 1×10^{10} to about 1×10^{12} GC per dose including all integers or fractional amounts within the range. All dosages may be measured by any known method, including as measured by oqPCR or digital droplet PCR (ddPCR) as described in, e.g., M. Lock et al, Hum Gene Ther Methods. 2014 April;25(2): 115-25. doi: 10.1089/hgtb.2013.131, which is incorporated herein by reference.

(75) These above doses may be administered in a variety of volumes of carrier, excipient or buffer formulation, ranging from about 25 to about 1000 microliters, including all numbers within the range, depending on the size of the area to be treated, the viral titer used, the route of administration, and the desired effect of the method. In one embodiment, the volume of carrier, excipient or buffer is at least about 25 μ L. In one embodiment, the volume is about 50 μ L. In another embodiment, the volume is about 75 μ L. In another embodiment, the volume is about 100 μ L. In another embodiment, the volume is about 125 μ L. In another embodiment, the volume is about 150 μ L. In another embodiment, the volume is about 175 μ L. In yet another embodiment, the volume is about 200 μ L. In another embodiment, the volume is about 225 μ L. In yet another embodiment, the volume is about 250 μ L. In yet another embodiment, the volume is about 275 μ L. In yet another embodiment, the volume is about 300 μ L. In yet another embodiment, the volume is about 325 μ L. In another embodiment, the volume is about 350 μ L. In another embodiment, the volume is about 375 μ L. In another embodiment, the volume is about 400 μ L. In another embodiment, the volume is about 450 μ L. In another embodiment, the volume is about 500 μ L. In another embodiment, the volume is about 550 μ L. In another embodiment, the volume is about 600 μ L. In another embodiment, the volume is about 650 μ L. In another embodiment, the volume is about 700 μ L. In another embodiment, the volume is between about 700 and 1000 μ L.

(76) In one embodiment, the viral constructs may be delivered in doses of from at least 1×10^9 to about 1×10^{11} GCs in volumes of about 1 μ L to about 3 μ L for small animal subjects, such as mice. For larger veterinary subjects having eyes about the same size as human eyes, the larger human dosages and volumes stated above are useful. See, e.g., Diehl et al, J. Applied

Toxicology, 21:15-23 (2001) for a discussion of good practices for administration of substances to various veterinary animals. This document is incorporated herein by reference.

(77) It is desirable that the lowest effective concentration of virus or other delivery vehicle be utilized in order to reduce the risk of undesirable effects, such as toxicity, retinal dysplasia and detachment. Still other dosages in these ranges may be selected by the attending physician, taking into account the physical state of the subject, preferably human, being treated, the age of the subject, the particular ocular disorder and the degree to which the disorder, if progressive, has developed.

(78) Yet another aspect described herein is a method for treating, retarding or halting progression of blindness in a mammalian subject having, or at risk of developing, choroideremia. In one embodiment, a rAAV carrying the REP-1 codon optimized sequences, preferably suspended in a physiologically compatible carrier, diluent, excipient and/or adjuvant, may be administered to a desired subject including a human subject. This method comprises administering to a subject in need thereof any of the nucleic acid sequences, expression cassettes, rAAV genomes, plasmids, vectors or rAAV vectors or compositions containing them. In one embodiment, the composition is delivered subretinally. In another embodiment, the composition is delivered intravitreally. In still another embodiment, the composition is delivered using a combination of administrative routes suitable for treatment of ocular diseases, and may also involve administration via the palpebral vein or other intravenous or conventional administration routes.

(79) Yet another aspect described herein is a method for treating, retarding or halting progression of blindness in a mammalian subject having, or at risk of developing, achromatopsia. In one embodiment, an rAAV carrying the CNGA3 or CNGB3 native, modified or codon optimized sequence, preferably suspended in a physiologically compatible carrier, diluent, excipient and/or adjuvant, may be administered to a desired subject including a human subject. This method comprises administering to a subject in need thereof any of the nucleic acid sequences, expression cassettes, rAAV genomes, plasmids, vectors or rAAV vectors or compositions containing them. In one embodiment, the composition is delivered subretinally. In another embodiment, the composition is delivered intravitreally. In still another embodiment, the composition is delivered using a combination of administrative routes suitable for treatment of ocular diseases, and may also involve administration via the palpebral vein or other intravenous or conventional administration routes.

(80) For use in these methods, the volume and viral titer of each dosage is determined individually, as further described herein, and may be the same or different from other treatments performed in the same, or contralateral, eye. The dosages, administrations and regimens may be determined by the attending physician given the teachings of this specification. In one embodiment, the composition is administered in a single dosage selected from those above listed in a single affected eye. In another embodiment, the composition is administered as a single dosage selected from those above listed in a both affected eyes, either simultaneously or sequentially. Sequential administration may imply a time gap of administration from one eye to another from intervals of minutes, hours, days, weeks or months. In another embodiment, the method involves administering the compositions to an eye two or more dosages (e.g., split dosages). In another embodiment, multiple injections are made in different portions of the same eye. In another embodiment, a second administration of an rAAV including the selected expression cassette (e.g., CHM containing cassette) is performed at a later time point. Such time point may be weeks, months or years following the first administration. Such second administration is, in one embodiment, performed with an rAAV having a different capsid than the rAAV from the first administration. In another embodiment, the rAAV from the first and second administration have the same capsid.

(81) In still other embodiments, the compositions described herein may be delivered in a single composition or multiple compositions. Optionally, two or more different AAV may be delivered, or multiple viruses [see, e.g., WO 2011/126808 and WO 2013/049493]. In another embodiment,

multiple viruses may contain different replication-defective viruses (e.g., AAV and adenovirus).

(82) In certain embodiments of the invention it is desirable to perform non-invasive retinal imaging and functional studies to identify areas of the rod and cone photoreceptors to be targeted for therapy. In these embodiments, clinical diagnostic tests are employed to determine the precise location(s) for one or more subretinal injection(s). These tests may include electroretinography (ERG), perimetry, topographical mapping of the layers of the retina and measurement of the thickness of its layers by means of confocal scanning laser ophthalmoscopy (cSLO) and optical coherence tomography (OCT), topographical mapping of cone density via adaptive optics (AO), functional eye exam, etc, depending upon the species of the subject being treated, their physical status and health and the dosage. In view of the imaging and functional studies, in some embodiments of the invention one or more injections are performed in the same eye in order to target different areas of the affected eye. The volume and viral titer of each injection is determined individually, as further described herein, and may be the same or different from other injections performed in the same, or contralateral, eye. In another embodiment, a single, larger volume injection is made in order to treat the entire eye. In one embodiment, the volume and concentration of the rAAV composition is selected so that only the region of damaged ocular cells is impacted. In another embodiment, the volume and/or concentration of the rAAV composition is a greater amount, in order reach larger portions of the eye, including non-damaged photoreceptors.

(83) In one embodiment of the methods described herein, a one-time intra-ocular delivery of a composition as described herein, e.g., an AAV delivery of an optimized REP-1 cassette, is useful in preventing vision loss and blindness in a subject at risk of developing choroideremia. In another embodiment of the methods described herein, a one-time intra-ocular delivery of a composition as described herein, e.g., an AAV delivery of an optimized CNGA3 or CNGB3 cassette, is useful in preventing vision loss and blindness in a subject at risk of developing achromatopsia.

(84) Thus, in one embodiment, the composition is administered before disease onset. In another embodiment, the composition is administered prior to the initiation of vision impairment or loss. In another embodiment, the composition is administered after initiation of vision impairment or loss. In yet another embodiment, the composition is administered when less than 90% of the rod and/or cones or photoreceptors are functioning or remaining, as compared to a non-diseased eye.

(85) In another embodiment, the method includes performing additional studies, e.g., functional and imaging studies to determine the efficacy of the treatment. For examination in animals, such tests include retinal and visual function assessment via electroretinograms (ERGs) looking at rod and cone photoreceptor function, optokinetic nystagmus, pupillometry, water maze testing, light-dark preference, optical coherence tomography (to measure thickness of various layers of the retina), histology (retinal thickness, rows of nuclei in the outer nuclear layer, immunofluorescence to document transgene expression, cone photoreceptor counting, staining of retinal sections with peanut agglutinin-which identifies cone photoreceptor sheaths).

(86) Specifically for human subjects, following administration of a dosage of a compositions described in this specification, the subject is tested for efficacy of treatment using electroretinograms (ERGs) to examine rod and cone photoreceptor function, pupillometry visual acuity, contrast sensitivity color vision testing, visual field testing (Humphrey visual fields/Goldmann visual fields), perimetry mobility test (obstacle course), and reading speed test. Other useful post-treatment efficacy test to which the subject is exposed following treatment with a pharmaceutical composition described herein are functional magnetic resonance imaging (fMRI), full-field light sensitivity testing, retinal structure studies including optical coherence tomography, fundus photography, fundus autofluorescence, adaptive optics laser scanning ophthalmoscopy, mobility testing, test of reading speed and accuracy, microperimetry and/or ophthalmoscopy. These and other efficacy tests are described in U.S. Pat. No. 8,147,823; in co-pending International patent application publication WO 2014/011210 or WO 2014/124282, incorporated by reference.

(87) In yet another embodiment, any of the above described methods is performed in combination

with another, or secondary, therapy. In still other embodiments, the methods of treatment of these ocular diseases involve treating the subject with the composition described in detail herein in combination with another therapy, such as antibiotic treatment, palliative treatment for pain, and the like. The additional therapy may be any now known, or as yet unknown, therapy which helps prevent, arrest or ameliorate these mutations or defects or any of the effects associated therewith. The secondary therapy can be administered before, concurrent with, or after administration of the compositions described above. In one embodiment, a secondary therapy involves non-specific approaches for maintaining the health of the retinal cells, such as administration of neurotrophic factors, anti-oxidants, anti-apoptotic agents. The non-specific approaches are achieved through injection of proteins, recombinant DNA, recombinant viral vectors, stem cells, fetal tissue, or genetically modified cells. The latter could include genetically modified cells that are encapsulated. (88) In one embodiment, a method of generating a recombinant rAAV comprises obtaining a plasmid containing an AAV expression cassette as described above and culturing a packaging cell carrying the plasmid in the presence of sufficient viral sequences to permit packaging of the AAV viral genome into an infectious AAV envelope or capsid. Specific methods of rAAV vector generation are described above and may be employed in generating a rAAV vector that can deliver the codon optimized REP-1 or CNGA3 or CNGB3 in the expression cassettes and genomes described above and in the examples below.

(89) In yet another embodiment, a vector comprising any of the expression cassettes described herein is provided. As described above, such vectors can be plasmids of variety of origins and are useful in certain embodiments for the generation of recombinant replication defective viruses as described further herein.

(90) In one another embodiment, the vector is a plasmid that comprises an expression cassette, wherein the expression cassette comprises AAV inverted terminal repeat sequences and a codon optimized nucleic acid sequence that encodes REP-1, and expression control sequences that direct expression of the encoded protein in a host cell.

(91) In another embodiment, the vector is a plasmid that comprises an expression cassette, wherein the expression cassette comprises AAV inverted terminal repeat sequences and a codon optimized nucleic acid sequence that encodes CNGA3, and expression control sequences that direct expression of the encoded protein in a host cell.

(92) In another embodiment, the vector is a plasmid that comprises an AAV expression cassette, wherein the expression cassette comprises AAV inverted terminal repeat sequences and a codon optimized nucleic acid sequence that encodes CNGB3, and expression control sequences that direct expression of the encoded protein in a host cell.

(93) It is to be noted that the term “a” or “an” refers to one or more. As such, the terms “a” (or “an”), “one or more,” and “at least one” are used interchangeably herein.

(94) The words “comprise”, “comprises”, and “comprising” are to be interpreted inclusively rather than exclusively. The words “consist”, “consisting”, and its variants, are to be interpreted exclusively, rather than inclusively. While various embodiments in the specification are presented using “comprising” language, under other circumstances, a related embodiment is also intended to be interpreted and described using “consisting of” or “consisting essentially of” language.

(95) As used herein, the term “about” means a variability of 10% from the reference given, unless otherwise specified.

(96) The term “regulation” or variations thereof as used herein refers to the ability of a composition to inhibit one or more components of a biological pathway.

(97) Unless defined otherwise in this specification, technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art and by reference to published texts, which provide one skilled in the art with a general guide to many of the terms used in the present application.

(98) The following examples are illustrative only and are not intended to limit the present

invention.

Example 1—Differentiation of Pluripotent Stem Cells into RPE

(99) Choroideremia lacks a relevant mouse model and there is no canine model, therefore, transduction and expression is tested in a human retinal cell model of the disease. Because it is impossible to obtain retinal cells from a living patient, RPE are generated from induced pluripotent stem cells. Pluripotent stem cells are directed to RPE using the protocol described by Buchholz et al, Rapid and Efficient Directed Differentiation of Human Pluripotent Stem Cells Into Retinal Pigmented Epithelium, Stem Cells Translational Medicine, 2013; 2:384-393 which is incorporated by reference in its entirety. See also, Cereso et al, Proof of concept for AAV2/5-mediated gene therapy in iPSC-derived retinal pigment epithelium of a choroideremia patient, Molecular Therapy—Methods & Clinical Development (2014) 1, 14011, which is incorporated by reference in its entirety. Other methods for producing RPE are known in the art.

(100) Briefly, the human induced pluripotent stem cell line is maintained in Dulbecco's Modified Eagle's Medium: Nutrient Mixture F-12 (DMEM/F12) containing 2mM GlutaMAX-I, 20% knockout serum replacement, 0.1 mM Modified Eagle's Medium Non-Essential Amino Acids (MEM NEAA), 0.1 mM β -mercaptoethanol and 4 ng/ml bFGF on a mitomycin C-treated or irradiated mouse embryonic fibroblast feeder layer.

(101) Pluripotent stem cells are passaged directly onto Matrigel (BD Biosciences) in DMEM/F12 with 1×B27, 1×N2, and 1×NEAA (Invitrogen). From days 0 to 2, 50 ng/ml Noggin, 10 ng/ml Dkk1, 10 ng/ml IGF1 and 10 mM nicotinamide are added to the base medium. From days 2 to 4, 10 ng/ml Noggin, 10 ng/ml Dkk1, 10 ng/ml IGF1, 5 ng/ml bFGF and 10mM nicotinamide are added to the base medium. From days 4 to 6, 10 ng/ml Dkk1, 10 ng/ml IGF1 and 100 ng/ml Activin A (R&D Systems) are added to the base medium. From days 6 to 14, 100 ng/ml Activin A, 10 μ M SU5402 (EMD Millipore, Darmstadt, Germany), and 1 mM VIP are added to the base medium. Control experiments are performed in base media alone (DMEM/F12, B27, N2, and NEAA).

(102) The cells are mechanically enriched by scraping away cells with non-RPE morphology. Subsequently, the remaining RPE are digested using TrypLE Express (Invitrogen) for 5 minutes at 37° C. The cells are passed through a 30- μ m single-cell strainer and seeded onto Matrigel-coated tissue culture plastic, Transwell membranes or CC2-treated chambered slides. Enriched cells are cultured in DMEM-high glucose with 1% fetal bovine serum (FBS), GlutaMAX, and sodium pyruvate for 30 days.

Example 2—Cells Transduced with AAV-REP-1

(103) Briefly, AAV2/8CMV.CBA-REP-1 viral vector incorporating REP-1 codon optimized sequences are produced by transient transfection of HEK293 cells, and the viral particles are precipitated from either the supernatant using polyethylene glycol. See, e.g., Guo et al, Rapid and simplified purification of recombinant adeno-associated virus, J Virol Methods. 2012 August; 183(2): 139-146, which is incorporated herein by reference. The vectors are purified by double CsCl centrifugation, dialyzed, and titered by dot blot assay.

(104) For the prenylation experiments, RPE are seeded in 24-well plates, and 1.2×10^6 cells are estimated at confluence. Cells are transduced with 100,000 vg per cell, and prenylation assays are performed at 4 weeks posttransduction. Experiments are performed in triplicate.

Example 3—Prenylation

(105) An in vitro prenylation assay is performed as described in Vasireddy et al, PLoS One. 2013 May 7;8(5): c61396, cited above, using [3H]-geranylgeranyl pyrophosphate (GGPP) (Perkin Elmer, Boston, MA, USA) as a prenyl group donor, in the presence of recombinant Rab geranylgeranyl transferease and RAB27. Incorporation of radiolabeled prenyl groups into the RAB27 protein is measured by scintillation counting. For consistency the control values are normalized to 100 and used as the base value. All experiments are performed in triplicate, and statistical comparison of prenylation between experimental and control groups is evaluated using the two-tailed unpaired

student's t-test.

(106) Briefly, 48 hr post transduction, transduced REP cells are washed with cold PBS. Cell pellets are collected and washed thoroughly with cold PBS. Cells are lysed on ice for 30 min using RIPA+Protease inhibitors. In an alternative protocol, cells are sonicated. Cytosolic fractions are collected by centrifuging the lysate at 75,000-100,000 g for 1-2 h at 4° C.

(107) Stocks are prepared for the prenylation reaction as follows.

(108) TABLE-US-00001 FINAL CONCENTRATION STOCKS prepared REQUIRED Rab GGTase 16.63 uM - (GGTase-a) 0.05 uM 14.59 uM - (GGTase-b) Rab 27a 25.93 uM 4 uM 3H GGPP 22.2 5 uM NP40 1 mM DTT 10 mM 1 mM HEPES 1M 50 mM Mgcl2 100 mM 5 mM

(109) Final reaction volume used for prenylation is 25 µL

(110) TABLE-US-00002 Rab GGTase a 0.075 ul Rab Ggtaseb 0.085 ul [.sup.3H]-geranylgeranyl 5.68 ul pyrophosphate (GGPP) NP40 0.15 ul DTT 2.5 ul HEPES 1.25 ul MgCl2 1.25 ul Rab 27a 3.12 ul Cytosolic Fraction (Cell lysate) 10.89 ul

(111) The reaction mixture is incubated at 37° C. for 30 min. The reaction is stopped by adding 9:1 ethanol/HCL, and incubated for 30 minutes. The proteins are collected on glass fiber filter papers (Whatman papers) by vacuum filtration (0.1 ml). The filters are washed carefully with cold phosphate buffer—3 times to remove unbound material. The membranes are dried carefully. The filters are placed in 5 ml scintillation cocktail and scintillation counting is performed. See also, Tolmachova et al, CHM/REP1 cDNA delivery by lentiviral vectors provides functional expression of the transgene in the retinal pigment epithelium of choroideremia mice, The Journal of Gene Medicine, 2012; 14-158-68, which is incorporated herein by reference in its entirety.

(112) Assays for CNGA3 or CNGB3 proof-of-concept may include use of a spontaneous mutant animal model (for example, the Cnga3^{-/-} mouse or the Awassi sheep). The mouse model could be bred with an “all-cone” photoreceptor mouse, the Nrl^{-/-} mouse, to obtain double knockouts. The latter (Cnga3^{-/-}Nrl^{-/-}) mouse may expedite identification of efficacy. Efficacy could be measured by pupillometry, measures of visual acuity and contrast (for example, using optokinetics), electroretinograms, and visual behavior. Ultimately, histology will document expression of the transgene with improved outcomes on the other measures. Hsitologic approaches will also be used to document any effects of the intervention on cone photoreceptors (total number of cone photoreceptors, density, location, etc).

(113) Similar to choroideremia as discussed above, assays for proof-of-concept for gene augmentation therapy for CNGA3- or CNGB3-associated achromatopsia may include use of induced pluripotent stem cell (iPSC) models. The iPSC models, generated from patients with achromatopsia due to CNGA3 or CNGB3 mutations, will be differentiated into retinal precursors and/or photoreceptor cells in vitro. The wildtype CNGA3 (or CNGB3) cDNA will be delivered to these cells using recombinant AAV and the cells will be analyzed for biogenesis and preservation of function of the relevant (Cyclic nucleotide-gated, CNG) channel comprised by these subunits. Channel function will be assessed by electrophysiology on membrane patches. Restoration of the channel should rescue cGMP-activated currents. Additional studies can test for sensitivity of channel function before and after delivery of the wildtype CNG cDNA to physiological ligands.

Example 4: In Vitro Expression of AAV.Codon-Optimized Human CHM

(114) The objective of this study was to evaluate the ability of AAV mediated CHM expression after gene delivery using a series of next generation AAV 2 and AAV8 vectors encoding the codon optimized CHM gene (SEQ ID NO: 1) in 84-31 and COS-7 cell lines.

(115) To maximize the expression of CHM, a codon optimized CHM sequence was produced (SEQ ID NO: 1). The codon optimized plasmid was synthesized and used in the creation of all the next generation CHM transgene expression cassettes. To overcome the potential problem of contamination of non-functional AAV genomes, a non-coding lambda stuffer region was included in the vector backbone. Incorporation of stuffer not only increases the length of the plasmid, but also diminishes the possibility of plasmid DNA backbone contamination while packaging the AAV.

The impact of incorporating a stuffer region in the vector backbone to eliminate the plasmid DNA impurities was carried out as an independent study. Two recombinant AAV proviral plasmids (high and low copy) backbones were used to generate the different constructs. The high copy plasmid was designed based on the pUC vector origin. The low copy plasmid was designed based on the p15A origin. To further enhance the translation from the correct initiation codon, a Kozak sequence upstream of the start codon was incorporated.

(116) A total of four plasmids have been engineered for the current study and those described in the following examples (Table 1). In addition, a plasmid carrying the CHM native sequence, which is currently being used in a clinical trial, was also generated (version 1). Plasmid maps for each of Version 2a, 2b, 3a and 3b, and Version 1 are shown in FIGS. 6-10, respectively.

(117) TABLE-US-00003

TABLE 1	Plasmid features	REP	Lambda	Kozak	Copy	Name	Sequence
insert	Sequence	Origin	number	Promoter/Intron	Version 1	Native	Present
absent	pUC	High	CMV-	CBA promoter + (V1)	copy	Enhancer	extension
Version 2a	Codon-	Present	present	pUC	High	CMV-CBA promoter + (V2a)	optimized copy
Version 2b	Codon-	Not	present	pUC	High	CMV-CBA promoter + (V2b)	optimized
Version 3a	Codon-	Present	present	p15A	Low	CMV-CBA promoter + (V3a)	optimized
Version 3b	Codon-	Not	present	p15A	Low	CMV-CBA promoter + (V3b)	optimized
Present	copy	Enhancer	extension				

(118) The in vitro expression of these constructs was tested in COS-7 and 84-31 cell lines. The engineered features of the next-generation CHM constructs are depicted in Table.1.

(119) Recombinant AAV proviral high and low copy plasmids were generated by cloning the codon optimized human CHM cDNA (hCHM) (SEQ ID NO: 1) into the transgene cassette. The transgene was placed under the control of a hybrid chicken β actin (CBA) promoter. This promoter consists of the cytomegalovirus (CMV) immediate early enhancer, the proximal chicken β actin promoter, and CBA exon 1 flanked by intron 1 sequences. The proviral high and low copy number plasmids also contain AAV inverted terminal repeats and a PolyA sequence. The next generation plasmid backbones used in the current study contain a lambda phage fragment stuffer followed by the kanamycin bacterial selection gene. Additional plasmids lack the stuffer but contain the kanamycin selection gene. The high-copy number vector is similar to that of pUC plasmids (~300 copies/bacterial cell). The low copy number plasmid (~10 copies/bacterial cell) has an origin of p15A. To enhance translation from the correct initiation codon, all next generation constructs contain a KOZAK consensus sequence upstream of the start codon, ATG. The generated plasmids are sequence verified using primers that can specifically target either the promoter+enhancer extension sequence or the codon optimized CHM sequence. The plasmid maps and sequences of all five constructs are shown in FIGS. 6-10.

(120) Standard triple transfection with calcium phosphate was used to generate AAV vectors listed below (see Table 2 for vector qualification). Both AAV2 and AAV8 serotypes of the vectors were generated to ensure the results are serotype-independent.

(121) TABLE-US-00004

TABLE 2	Summary of AAV2 and AAV8 vectors generated and concentration of viral stocks	Stock Conc.	Name	Serotype	Plasmid name	Lot number (vg/ml)
AAV2.V1	AAV 2 Version 1 (V1)	KA 892*	4.47E+12	AAV2.V2a	AAV 2 Version 2a (V2a)	CT 239
2.16E+12	AAV2.V2b	AAV 2 Version 2b (V2b)	CT 238	7.40E+12	AAV2.V3a	AAV 2 Version 3a (V3a)
CT 258	4.82E+12	AAV2.V3b	AAV 2 Version 3b (V3b)	CT 256	5.91E+12	AAV8.V1
AAV 8 Version 1 (V1)	KA 808*	1.39E+13	AAV8.V2a	AAV 8 Version 2a (V2a)	CT 245	1.04E+13
AAV8.V2b	AAV 8 Version 2b (V2b)	CT 244	1.11E+13	AAV8.V3a	AAV 8 Version 3a (V3a)	CT
259	8.67E+12	AAV8.V3b	AAV 8 Version 3b (V3b)	CT 255	1.36E+13	

(122) The 84-31 cell line is a subclone of 293 HEK cell line (human embryonic kidney cells) and constitutively expresses adenovirus E4 proteins to enhance transduction of AAV virus. COS-7 cells are fibroblast like cell lines that are derived from monkey kidney tissues. Both 84-31 cells and COS-7 cells were plated, separately, in 6-well cell culture plates and transduced with one of the ten

test articles (either AAV2 or AAV8) at five different multiplicity of infection (MOIs). After 36-48 hours, cells were harvested, lysed and protein samples were prepared for SDS-PAGE followed by Western blot analysis to detect the expression of exogenous CHM.

(123) Both 84-31 and COS-7 cells were cultured in Dulbecco's modified Eagle medium (DMEM)-high glucose with 10% fetal bovine serum, and 1% penicillin/streptomycin at 37° C. in an environment supplied with 5% CO₂. The day before transduction (18-24 h before) cells at a density of 3E5 were seeded in 2 ml of cell culture media in each well of a 6-well cell culture dish. Seeded cells were incubated at 37° C. in an environment supplied with 5% CO₂. Wells of both COS-7 and 84-31 cells were infected with AAV vectors listed below at various multiplicities of infection (MOI) (Table 3 and Table 4). No virus was added to negative control cells (untransduced cells). Briefly, the tissue culture media was removed and a fresh 2 ml aliquot of media was added to each well of the 6 well culture dish. Then the predetermined amount of AAV vector was measured (directly from the stock) and added to each well (Table 3 and Table 4). For an MOI of 1E4, 1 µL of respective virus stock was diluted to 10 µL with cell culture media. From this solution, the predetermined volume of the virus was added to respective well (Table 3 and 4). Cells were incubated with the AAV virus for 36-48 hours at 37° C. with 5% CO₂ till harvesting. Cells were observed under microscope before harvesting to check for abnormalities.

(124) TABLE-US-00005 TABLE 3 Infection dose of four next generation AAV2 and AAV8 hCHM vectors in COS-7 cells. Vector Used Vector Used Cell Line Cell Density (µL) MOI No AAV COS-7 3E5 — — AAV2.V2a COS-7 3E5 1.5 1E4 COS-7 3E5 15 1E5 COS-7 3E5 45 3E5 COS-7 3E5 75 5E5 COS-7 3E5 150 1E6 AAV2.V2b COS-7 3E5 4.2 (front a 1 to 1E4 10 dilution of the stock) COS-7 3E5 4.2 1E5 COS-7 3E5 12.6 3E5 COS-7 3E5 21 5E5 COS-7 3E5 42 1E6 AAV8.V2a COS-7 3E5 2.9 (from a 1 to 1E4 10 dilution of the stock) COS-7 3E5 2.88 1E5 COS-7 3E5 8.65 3E5 COS-7 3E5 14.42 5E5 COS-7 3E5 28.85 1E6 AAV8.V2b COS-7 3E5 2.7 (front a 1 to 1E4 lb dilution of the stock) COS-7 3E5 2.7 1E5 COS-7 3E5 8.1 3E5 COS-7 3E5 13.5 5E5 COS-7 3E5 27 1E6

(125) TABLE-US-00006 TABLE 4 Infection rates of four next generation AAV2 and AAV8 hCHM vectors in 84-31 cells. Vector Used Vector Used Cell Line Cell Density (µL) MOI No AAV 84-31 3E5 — — AAV2.V2a 84-31 3E5 1.5 1E4 84-31 3E5 15 1E5 84-31 3E5 45 3E5 84-31 3E5 75 5E5 84-31 3E5 150 1E6 AAV2.V2b 84-31 3E5 4.2 (from a 1 to 1E4 10 dilution of the stock) 84-31 3E5 4.2 1E5 84-31 3E5 12.6 3E5 84-31 3E5 21 5E5 84-31 3E5 42 1E6 AAV8.V2a 84-31 3E5 2.9 (from a 1 to 1E4 10 dilution of the stock) 84-31 3E5 2.88 1E5 84-31 3E5 8.65 3E5 84-31 3E5 14.42 5E5 84-31 3E5 28.85 1E6 AAV8.V2b 84-31 3E5 2.7 (from a 1 to 1E4 10 dilution of the stock) 84-31 3E5 2.7 1E5 84-31 3E5 8.1 3E5 84-31 3E5 13.5 5E5 84-31 3E5 27 1E6

(126) First, both, the COS7 and 84-31 cell lines were used to test if the in vitro expression of CHM is cell line independent. Once the independence was established, all subsequent experiments were carried out only in 84-31 cells, which have shown superior transduction efficiency with AAV. Wells of 84-31 cells were infected with the AAV vectors listed below at various MOI (see table 3 and 4).

(127) Western blot analysis: 1. Cell lysates were prepared. The AAV transduced cells along with the untreated control cells, were harvested 36-48 h post-infection after a thorough PBS wash. Cells were then lysed on ice using RIPA buffer with protease inhibitors. Cell lysates were cleared by centrifuging at 13,000 rpm for 10 min. 2. Quantification and preparation of proteins. Protein quantification of the cell lysates was carried out using ThermoFisher Micro BCA™ Protein Assay Kit following manufacturer's instructions. Protein concentration was determined by taking OD reading at 562 nm. To evaluate the in vitro expression of CHM, between 40-60 ug of measured protein was loaded on 4-12% Bis-Tris gels. 3. SDS-PAGE and blotting SDS-PAGE and western blot analysis were carried out according to known protocols. Briefly, the protein gels were transferred on to a nitrocellulose membrane, blocked in milk and incubated with the primary antibodies. Antihuman REP-1 2F1 antibody (2F1, 1:1000 dilution) and one of the following: anti-GAPDH antibody (1:1000 dilution), anti actin antibody (1:1000 dilution) or anti-Tubulin antibody

(1:5000 dilution) was used as primary antibodies for each blot. After washing the blot, HRP conjugated anti-mouse IgG antibody and/or anti-rabbit IgG antibody at a concentration of 1:5000 were used as secondary antibodies. The blots were developed by chemiluminescence using ECL reagents according to the manufacturer's instructions. Controls: 1. Loading controls: One of the following: anti-Actin antibody, anti-tubulin antibody or anti-GAPDH antibody was used as a loading control to demonstrate equal loading of protein in each well of the gels. Anti-Tubulin antibody detects a protein of ~51 kDa. Anti-Actin antibody detects a protein of ~42 kDa, and anti-GAPDH antibody detects a protein of ~39 kDa. Initial blots were probed with either anti-tubulin antibody or anti-Actin antibody or anti-GAPDH antibody depending up on their availability. After initial experiments, to be consistent, anti-GAPDH antibody was used as the loading control. 2. Positive control: After the production of hREP-1 protein was established in AAV2.V2a transduced COS-7 cells, the AAV2.V2a-Cos-7 cell lysates were used as positive control in later western blot experiments. 3. Negative control: Untreated cells were used as negative control. Analyses of western blot results of REP-1 protein production in various cell lines are summarized in Table. 5. (128) TABLE-US-00007 TABLE 5 CHM Expression Name Serotype Cell Line FIG. MOI Used (Observation) AAV2.V2a AAV2 COS-7 FIG. 1 1E4-1E6 Detectable AAV2.V2a AAV2 84-31 FIG. 1 1E4-1E6 expression of AAV2.V2b AAV2 COS-7 FIG. 2 1E4-1E6 CHM at all AAV2.V2b AAV2 84-31 FIG. 2 1E4-1E6 MOIs tested AAV2.V3a AAV2 84-31 FIG. 3 1E4-1E6 AAV2.V3b AAV2 84-31 FIG. 3 1E4-1E6 AAV2.V2a AAV8 COS-7 FIG. 4 1E4-1E6 Detectable AAV2.V2a AAV8 84-31 FIG. 4 1E4-1E6 expression of CHM at MOI of 1E5-1E6. AAV2.V2b AAV8 COS-7 FIG. 5 1E4-1E6 Detectable AAV2.V2b AAV8 84-31 FIG. 5 1E4-1E6 expression of CHM above MOI of 3E5-1E6 AAV2.V3a AAV8 84-31 FIG. 6 1E4-1E6 Detectable AAV8.V3b AAV8 84-31 FIG. 6 1E4-1E6 expression of CHM above MOI of 3E5-1E6

(129) Monoclonal human REP-1-specific antibody, detected one single ~75-80 kDa hREP-1 protein in cells transduced with next generation AAV2.copt.CHM/AAV8.copt.CHM. A 75-80 kDa band was not observed in cell lysates of untreated control cells. Probing of the blots with either anti-Actin/antitubulin/anti-GAPDH antibody showed a band of equal density in all lanes of the western blot including in untreated controls. Anti-actin antibody detected a protein molecular weight band at ~42 kDa, anti-tubulin antibody detected a protein at ~51 kDa, and anti-GAPDH antibody detected a protein at ~39 kDa. All antibodies detected only specific bands of expected size molecular weight. No nonspecific bands were observed in any of the blots. A pre-stained molecular weight marker was used to compare the molecular weights of protein of interest.

(130) Briefly, REP-1 protein was observed at the expected size in COS-7 and 84-31 cells transduced with AAV2.V2a, AAV2.V2b, AAV2.hCHM.V3a and AAV2.hCHM.V3b. Untreated controls did not reveal the presence of expected size human REP-1 protein. Labeling the blot with anti-actin antibody detected a protein band of equal intensity in all lanes of the gel at ~42 kDa. Pre-stained protein ladder was used to compare the molecular weights of REP-1 and Actin. Data not shown.

(131) The results indicate that AAV2 and AAV8 serotype vectors containing next generation plasmids are able to transduce 84-31 and COS-7 cells efficiently. Expression of CHM in the next generation plasmids was in the detectable range, and demonstrated a dose dependent trend. Transduction of cells with the next generation hCHM viruses resulted in production of REP-1 protein of the predicted size.

Example 5: Comparison of in Vitro Protein Expression of AAV.Codon-Optimized.Human CHM with AAV Native.Human CHM

(132) The objective of this study was to delineate transduction efficiency of AAV vectors (serotype 2 and 8) containing various versions of the CHM-containing transgene cassettes by measuring levels of REP-1 protein in a 84-31 cell line based study model.

(133) Plasmids and Vectors: A total of 5 transgene plasmids were compared either in AAV2 or AAV8: Version 1 (previously being used in an on-going clinical trial) and four next generation

versions (V2a, V2b, V3a, and V3b). The plasmids were engineered as described in Example 4, and the features thereof are shown in Table 1. Table 2 above shows a summary of AAV2 and AAV8 vectors generated and concentration of viral stocks.

(134) Study Design (e.g. Treatment Groups)

(135) 1. In a pilot experiment, COS-7 and 84-31 cells were transduced with AAV2.hCHM.Version1, Version2a and Version 2b. Western blot was performed to compare transduction efficiency levels in the two cells lines.

(136) 2. 84-31 cells, plated in 6-well plates were transduced with one of the 10 test articles (Version 1, 2a, 2b, 3a and 3b in either AAV2 or AAV8 background) at an MOI of 3E5. After 36-48 hours, cells were harvested and lysed. The lysate was loaded on SDS-PAGE, and subjected to further Western blot analyses. Levels of REP-1 protein are compared amongst all construct versions. Two separate plates were setup for each AAV2.CHM or AAV8.CHM experiments were analyzed, separately.

(137) Test Material Administration

(138) 3.4.1 Cell Culture

(139) 83-41 cells and COS-7, both were cultured in Dulbecco's modified Eagle medium (DMEM)-high glucose with 10% fetal bovine serum, and 1% penicillin/streptomycin at 37° C. in an environment supplied with 5% CO2.

(140) 3.4.2 Preparation of Cells for Transduction:

(141) The day before transduction (18-24 h before) 83-41 and COS-7 cells were seeded at a density of 3E5 in 2 ml of cell culture media per well in a 6-well cell culture dish. The seeded cells were incubated at 37° C. in an environment supplied with 5% CO2.

(142) 3.4.3 Transduction:

(143) Wells of 84-31 cells and Cos-7 were infected with AAV vectors as described below at an MOI of 3E5 (see Table 6 for the pilot experiment and Table 7 for the second set of experiments). No virus was added to the negative (untransduced) control. Briefly, first, the tissue culture media was removed and replaced with 2 ml fresh media/well in each the wells in the 6 well cell culture dish. Then the predetermined amount of AAV vector (see table 2 for vector volumes used for transduction) was measured (from the stock) and directly added to each well. Cells were incubated with the AAV virus for 36-48 hours at 37° C. with 5% CO2 until harvesting. Cells were observed under microscope before harvesting to check any abnormality. Western blot analysis was performed as described in Example 4.

(144) TABLE-US-00008 TABLE 6 Pilot Experiment: Infection doses of AAV2.hCHM.V1, 2a, 2b in 84-31 and COS-7 cells. Vector Used Cell Line Cell density Vector used (μL) MOI None 84-31 3E5 0 0 AAV2.V1 84-31 3E5 13.42 3E5 AAV2.V2b 84-31 3E5 8.11 3E5 AAV2.V2a 84-31 3E5 27.78 3E5 AAV2.V1 COS-7 3E5 13.42 3E5 AAV2.V2b COS-7 3E5 8.11 3E5 AAV2.V2a COS-7 3E5 27.78 3E5 None COS-7 3E5 0 0

(145) TABLE-US-00009 TABLE 7 Infection doses of AAV.hCHM next generation vectors and V1 (AAV2 and AAV8) in 84-31 cells Vector Used Cell Line Cell density Vector used (μL) MOI None 84-31 3E5 0 0 AAV8.V2a 84-31 3E5 5.77 3E5 AAV8.V2b 84-31 3E5 5.41 3E5 AAV8.V3a 84-31 3E5 6.92 3E5 AAV8.V3b 84-31 3E5 4.41 3E5 AAV8.V1 84-31 3E5 4.32 3E5 None 84-31 3E5 0 0 AAV2.V2a 84-31 3E5 27.78 3E5 AAV2.V2b 84-31 3E5 8.11 3E5 AAV2.V3a 84-31 3E5 12.45 3E5 AAV2.V3b 84-31 3E5 10.15 3E5 AAV2.V1 84-31 3E5 13.42 3E5

Results: Comparison of the Expression of Native hCHM (AAV2.hCHM.V1) Versus Codon-Optimized CHM AAV2a and 2b Vectors in 84-31 and COS-7 Cells

(146) In this experiment 84-31 and COS-7 cells were transduced with either no vector (untreated control), AAV2.hCHM.Version1, AAV2.hCHM.Version2a or AAV2.hCHM.Version2b. Western blot analysis with an anti-human REP-1 antibody, showed that REP-1 protein levels were detectable at ~75-80 kDa in all AAV2 (V1, V2a, V2b) transduced samples and in both cells lines (Data not shown). A slightly better protein expression was seen in 84-31 cell line (Table 8). Anti-REP1

antibody detected negligible amount of REP-1 protein in untreated cells. Labeling of the blot with GAPDH antibody detected a band at ~39 kDa in all cell lysates, including the untreated cells. (147) Densitometric evaluation (quantification of the protein level) of the blots using ImageJ software demonstrated that after normalizing the values to the expression of endogenous GAPDH protein the transduction efficiency was similar in 84-31 and COS-7 cells. (See Table 8 for results.) Based on this, the 84-31 cell line, which is from human origin was used for further experiments. (148) In conclusion, AAV2.V1, AAV2.V2a and AAV2.V2b induced the production of REP-1 protein in both, 84-31 and COS-7 cells with similar transduction efficiency.

(149) TABLE-US-00010 GAPDH REP-1 REP-1 NORMALIZED NORMALIZED
NORMALIZED TO GAPDH OF TO RESPECTIVE TO REP-1 OF V1 REP1 GAPDH V1
GAPDH (%) 84-31 AAV2.V1 23416.844 19350.773 1 23416.844 100 AAV2.V2b 36626.765
20357.894 1.011 36203.838 154.605 AAV2.V2a 31114.844 20315.945 1.009 30819.0684 131.610
Molecular weight Marker COS7 AAV2.V1 12880.459 15479.288 1 12880.459 100 AAV2.V2b
19209.823 14321.167 0.711 26991.925 209.557 AAV2.V2a 15132.602 13145.924 0.849 17818.595
138.338

(150) Comparison of the expression of native CHM versus codon-optimized CHM AAV2 vectors in 84-31 cells: Using an anti-human REP-1 antibody, the Western blot analysis of the 84-31 cells transduced with AAV2.hCHM.V2a, V3a, V2b, V3b and V1 detected a band at ~75-80 kDa in all conditions (Data not shown). Anti-REP1 antibody detected negligible amount of REP-1 protein in untreated cells. Labeling of the blot with GAPDH antibody detected a band at ~39 kDa in all cell lysates, including the untreated cells. Densitometric evaluation (quantification of the expression level) of the blots using ImageJ software demonstrated an increase in the expression of AAV2.hCHM.V2a, 3a, 2b, and 3b compared to AAV2.hCHM.V1 after normalizing the values to the production of endogenous GAPDH protein. See Table 9 and 10 for results.

(151) TABLE-US-00011 TABLE 9 Values of REP-1 protein in 84-31 cells after transduction with AAV2.hCHM. V1, Va, V2b, V3a or V3b for PLATE 1 (Western Blot not shown) REP-1 GAPDH
NORMALIZED REP-1 CONSTRUCT NORMALIZED TO NORMALIZED LANE RAW VALUE
TO GAPDH OF RESPECTIVE TO REP-1 OF V1 NAME NUMBER REP-1 GAPDH V1 GAPDH
(%) AAV2.V1 1 23367.593 15155.602 1 23367.593 100 AAV2.V2a 3 26949.421 10969.581
0.723797115 37233.39103 159.3377248 AAV2.V3a 5 29867.714 14595.894 0.963069233
31013.0497 132.7182038 AAV2.V2b 7 32728.128 14133.551 0.932562824 35094.82381
150.1858741 AAV2.V3b 9 33986.543 13670.066 0.901981063 37679.88531 161.2484662

(152) TABLE-US-00012 TABLE 10 Values of REP-1 protein in 84-31 cells after transduction with AAV2.hCHM. V1, V2a, V2b, V3a or V3b for PLATE 2 (Western blot not shown) REP-1 GAPDH
NORMALIZED REP-1 CONSTRUCT NORMALIZED TO NORMALIZED LANE RAW VALUE
TO GAPDH OF RESPECTIVE TO REP-1 NAME NUMBER REP-1 GAPDH V1 GAPDH OF V1
(%) AAV2.V1 2 23128.593 11993.823 1 23128.593 100 AAV2.V2a 4 23623.836 10982.798
0.915704526 25798.53582 111.5439051 AAV2.V3a 6 28832.543 13176.359 1.098595419
26244.91465 113.473892 AAV2.V2b 8 31349.229 16028.329 1.336381986 23458.28463
101.4254721 AAV2.V3b 10 33273.856 14760 1.230633469 27037.99047 116.9028763

(153) Comparison of the expression of native CHM versus codon-optimized CHM in AAV8.V1, V2a, V3a, V2b, V3b vectors in 84-31 cells: Western blot analysis of cells transduced with AAV8.V1, AAV8.V2a, AAV8.V3a, AAV8.2b, AAV8.3b, with anti-human REP-1 antibody detected a band at ~75-80 kDa in all transduced cells (Data not shown). Anti-REP1 antibody detected negligible amount of REP-1 protein in untreated cells. Labeling of the blot with GAPDH antibody detected a band at ~39 kDa in all cell lysates, including the untreated cells. Densitometric evaluation of the blots using ImageJ software demonstrated higher expression of AAV8.hCHM.V2a; 3a; 2b; 3b compared to AAV8.V1. Values are obtained after normalizing the CHM values first, to the expression of the respective endogenous GAPDH protein and then normalized to the expression level of the average of Version 1. See Table 11 and Table 12 for

results.

(154) TABLE-US-00013 TABLE 11 Values of REP-1 protein expression in 84-31 cells after transduction with AAV8 hCHM Version 1, 2a, 2b, 3a and 3b - PLATE 1 (Western blot not shown)
GAPDH REP-1 NORMALIZED REP-1 CONSTRUCT TO TO NORMALIZED
LANE RAW VALUE GAPDH OF RESPECTIVE TO REP-1 OF NAME NUMBER REP-1
GAPDH V1 GAPDH V1 (%) AAV8.V1 11 3630.589 20309.924 1 3630.589 100 AAV8.V2a 13
7133.439 17051.48 0.839 8496.599 234.028 AAV8.V3a 15 5828.418 15801.045 0.777 7491.575
206.346 AAV8.V2b 17 11411.702 19249.681 0.947 12040.241 331.633 AAV8.V3b 19 17610.066
18727.024 0.922 19098.555 526.045

(155) TABLE-US-00014 TABLE 12 Values of REP-1 protein expression in 84-31 cells after transduction with AAV8 hCHM Version 1, 2a, 2b, 3a and 3b - PLATE 2 (Western blot not shown)
REP-1 GAPDH NORMALIZED REP-1 CONSTRUCT NORMALIZED TO NORMALIZED
LANE RAW VALUE TO GAPDH RESPECTIVE TO REP-1 OF NAME NUMBER REP-1
GAPDH OF V1 GAPDH V1 (%) AAV8.V1 12 3507.468 19082.681 1 3507.468 100 AAV8.V2a
14 4359.296 13274.731 0.695642871 6266.571801 178.6636913 AAV8.V3a 16 6533.246
20720.246 1.0858142 6016.909708 171.5456765 AAV8.V2b 18 13962.045 17842.167
0.934992677 14932.7854 425.7425983 AAV8.V3b 20 16049.823 13836.368 0.725074637
22135.40811 631.0936581

(156) Conclusion: Comparative expression studies demonstrated that application of AAV vectors carrying the next generation AAV.hCHM. Version 2a, 2b, 3a and 3b induced increased production of REP-1 protein compared with Version 1 (currently used in clinical trials) in both AAV2 and AAV8 serotype vectors in 84-31 cells.

Example 6: Evaluation of Lambda Stuffer's Effect on AAV Vector Production by qPCR Titer Analysis

(157) A single qPCR (quantitative polymerase chain reaction) run was performed on all 8 AAV vectors shown in Table 2, above, in order to assess the effect of lambda stuffer sequences on the amount of DNA impurities. Linearized AAV plasmid standard was used to generate the assay standard. Primer-probe sets were designed on either the CMV/CBA promoter region for quantitation of properly packaged AAV genome or the Kanamycin resistance (KanR) encoding region for reverse packaging. Standards and vector samples were run in two sets, one with the CMV/CBA primer-probe set and the other with the KanR set. Vector sample values (viral genome copy per mL) were determined from each respective standard curve. The effect of the stuffer sequence was assessed by comparing the relative amount of KanR-containing impurities in each vector lot against CMV/CBA containing sequences.

(158) Reagents:

(159) Transgene-Containing Viral Vector Titer:

(160) TABLE-US-00015 Reference: CMV-CBA Promoter Primers: CMV-F: CCC ACT
TGG CAG TAC ATC AACMV-R: GCC AAG TAG GAA AGT CCC ATA
A FAM-Probe: /56-FAM/CA TAA TGC C/ZEN/A GGC GGG CCA TTT
AC/3IABkFQ/

Impurity-Containing Viral Vector Titer

Reference: Kanamycin Resistance Gene

(161) TABLE-US-00016 Primers: KAN-F: GAT GGT CGG AAG TGG CAT AA
KAN-R: TGC GCC AGA GTT GTT TCT FAM-Probe: /56-FAM/CC GTC
AGC C/ZEN/A GTT TAG TCT GAC CA/3IABkFQ/

Dilution Reagent: Diluent Q (0.001% PF-68 in nuclease free water): Diluted 1% PF-68 solution
1000-folds with sterile water. Diluent S: Diluent Q+2 ng/μL salmon sperm DNA (Agilent
technologies Cat #201190)

ABI TaqMan™ Universal Master Mix (Applied Biosystems 4304437/4326708)

Qiagen PCR Product Purification Kit (Qiagen 28104).Math. ABI QuantStudio 6 Flex System

1.2. Sample Preparation

DNase digest solution was prepared by combining the following: DNase buffer (10×) 5 µL, Nuclease-free H₂O 30 µL, DNase I (Invitrogen, 18068-015) 5 µL

(162) Ten µL of each AAV vector sample was mixed in and incubated at ambient temperature for 10 minutes. The digest mix was inactivated by adding 50 µL of SDS/EDTA/NaCl solution (0.2% SDS/5 mM EDTA/0.2M NaCl) and incubating at 95° C. for 10 minutes. Each AAV vector sample was diluted 10-100,000 fold in Diluent S for qPCR analysis.

(163) 1.3. qPCR Standard Preparation

(164) Reference standard DNA (linearized) was prepared by digesting plasmid p1008 (low copy transgene plasmid without stuffer) with XhoI and purifying with Qiagen PCR purification kit. Purified material was analyzed on agarose gel to confirm identity, and quantified by Nanodrop. DNA copy number was determined from the stock concentration using the following equivalence: 1 bp=1.096E-21 g. qPCR Standards were prepared according to the following table:

(165) TABLE-US-00017 TABLE 13 Concentration DNA Standard [Copies/10 µL] Dilution
Standard Stock 2 × 10⁹ NA S1 1 × 10⁸ 10 µL Stock + 190 µL Diluent S S2 1 × 10⁷
10 µL S1 + 90 µL Diluent S S3 1 × 10⁶ 10 µL S2 + 90 µL Diluent S S4 1 × 10⁵ 10 µL S3
+ 90 µL Diluent S S5 1 × 10⁴ 10 µL S4 + 90 µL Diluent S S6 1 × 10³ 10 µL S5 + 90 µL
Diluent S

PCR Reaction Setup

(166) Extracted DNA samples were analyzed in triplicate (3 wells) in a single qPCR run. The run included reference DNA standards in triplicate, ranging from 103 to 108 copies per well. No-template-control (NTC) was included as negative control. Each AAV vector preparation was analyzed with both CMV/CBA and KanR primer/probe sets. Similarly, for quantitation of each set, the standards were also analyzed with both CMV/CBA and KanR primer/probe sets.

(167) TABLE-US-00018 TABLE 14 PCR Reaction Setup Final conc. in Volume per 25 Reagents
Reaction µL Reaction Universal Master Mix (2x) 1x 12.5 µL Optimized Primer Mix (20 µM) 0.8
µM 1.0 µL Taqman Probe (10 µM) 0.2 µM 0.5 µL Nuclease-Free Water NA 1.0 µL
Sample/Standard NA 10.0 µL

(168) PCR reaction sequence was set up as follows: 50° C. 2 minutes 1 cycle; 95° C. 10 minutes 1 cycle; 95° C. 15 seconds 40 cycles; 60° C. 1 minutes 40 cycles

(169) Run performance. Standards were prepared and run at 103 to 108 DNA copies per well. Lower limit of the assay was set at 1000 copies since assay sensitivity was not an important factor for this experiment. A standard curve was generated for the run using the standard copy numbers and CT (threshold cycle) values of the standards. Linear regression of the standards was performed using the ABI software (data not shown). Standard curve fit had a correlation coefficient (R² value) of 0.998 or greater indicating a reliable fit model. The slope of the standard curves was -3.5. Slope was used to calculate the efficiency of the amplification reaction, and values between -3.2 and -3.6 represented amplification efficiency between 90% and 110%. Both standard reactions were run with 92.6~93.8% efficiency. Precision of triplicate wells ranged from 2~10%, indicating good agreement among replicates. No-template-control (NTC) resulted in non-quantifiable amplification below the lower limit of the assay.

(170) TABLE-US-00019 TABLE 15 Summary of standard curve fit Reporter Target Slope Y-intercept R Efficiency (%) FAM CMV -3.513 41.896 0.998 92.597 FAM KAN -3.481 39.968 1.000 93.761

Results:

(171) Sample value determination: The sample values (AAV genome and reverse-packaging copy number) were interpolated from each matching standard curve (CMV/CBA or KanR), using CT values. Interpolated DNA copy number was corrected for initial dilution and/or digest dilution. Additional correction factor of 2 was applied to account for the difference between double-stranded DNA standards and single-stranded DNA in samples.

(172) Analysis results for 8 AAV vectors are summarized in the table below, with quantitative comparison between the transgene-containing AAV concentration (CMV/CBA) and the KanR-containing impurity concentration. Analysis of results demonstrate that insertion of lambda stuffer into the transgene plasmid effectively reduced the occurrence of plasmid-backbone DNA (i.e. KanR) packaging during AAV production from ~7-20 folds (FIG. 11).

(173) TABLE-US-00020 TABLE 16 qPCR amplification of kanamycin versus CMV/CBA expressed as percentage Kan vs. Lambda CMV/CBA qPCR KanR qPCR CMV/CBA Sample Name Stuffer (vg/ml) (vg/ml) (%) AAV2.V2b No 1.23E+13 6.46E+11 5.25% AAV2.V2a Yes 4.61E+12 3.60E+10 0.78% AAV8.V2b No 3.19E+13 1.43E+12 4.48% AAV8.V2a Yes 2.90E+13 1.19E+11 0.41% AAV2.V3b No 1.26E+13 6.69E+11 5.31% AAV2.V3a Yes 6.33E+12 4.56E+10 0.72% AAV8.V3b No 5.19E+13 3.07E+12 5.92% AAV8.V3a Yes 2.60E+13 8.00E+10 0.31%

Example 6: In Vitro Expression of Next Generation AAV8 Vectors in iPS Cells by Western Blot

(174) The objective of this study was to evaluate the ability of AAV mediated CHM expression after gene delivery using a series of next generation AAV2 and AAV8 vectors carrying the codon optimized REP-1-encoding gene in induced pluripotent cell lines (iPSC).

(175) Induced pluripotent stem (iPS) cell technology has been successfully utilized as a platform for testing gene therapy vectors in several proof-of-concept and gene therapy studies including ocular diseases. These patient-specific iPS cells provide a valuable in vitro model system to study disease pathogenesis and establish a model to test proof-of-concept of gene therapy where relevant animal models are unavailable. As a preliminary step to test our AAV-mediated gene augmentation therapy for Choroideremia (CHM), we have generated iPS cells from human patients harboring mutations in the causative gene, CHM, which encodes Rab Escort Protein 1 (REP-1) (See example 1) (Method is described in NCP.003). The generated iPS cells were used to evaluate the in vitro expression of our next generation AAV.codon optimized.CHM constructs.

(176) Plasmids and vectors were as described in Example 4. Induced pluripotent stem (iPS) cells are stem cells generated in the laboratory from somatic cells, peripheral blood mononuclear cells, that were reprogrammed back to a pluripotent state. Reprogramming of blood cells enables the development of personalized in vitro cellular models for therapeutic applications. In this report, iPS cells from individuals affected by CHM were used to test the in vitro production of REP-1 protein through western blot analysis. The following table (Table 17) describes the details of iPS cells studied and their respective CHM disease-causing mutations. Table 17: An overview of the iPS cells generated from patients with CHM mutations

(177) TABLE-US-00021 Cell Method of iPSC Line Affected Mutation in CHM generation JB 588 Affected Arg 555 stop Sendai virus mediated reprogramming JB Affected Exon 2-4 deletion Sendai virus mediated 527* reprogramming JB Affected Ex 10 c.1327_1327 del AT Sendai virus mediated 500* (Needs confirmation) reprogramming *iPS cell line qualification tests are on-going.

Study Design (e.g. Treatment Groups)

(178) 1. iPS cells plated on a 12 well cell culture plate are infected with AAV2. hCHM Version 1, Version 2a; Version 2b; Version 3a; Version 3b (AAV2.V1; V2a; V2b; V3a; V3b) at an MOI of either 1E5 or 3E5. After 24 hours of transduction, 1 ml of iPS cell culture media was added to the cells. 36-48 hours of transduction, cells were harvested, lysed and processed for SDS-PAGE followed by Western blot analysis. Production of REP-1 protein was evaluated in cells transduced with all versions of the constructs and compared with untreated controls.

(179) 2. As a pilot experiment, three different iPS cell lines plated on a 12 well cell culture plate are transduced with AAV8.hCHM Version 1 and AAV8.hCHM Version 2a (AAV8.V1; AAV8.V2a) at an MOI of 1E6. The iPS cell lines were derived from three CHM affected individuals with unrelated mutations in REP1 gene and were plated in separate plates for this purpose. After 36-48 hours, cells were harvested and lysed and subjected to Western blot analyses compared with untreated cell lysate.

(180) Test Material Administration

(181) 3.4.1 Cell Culture

(182) Culturing of iPS cells from CHM patient. In brief, the iPS cells were cultured on Mouse Embryonic Fibroblasts (MEFs, feeders) in iPS cell culture media at 37° C. in an environment supplied with 5% CO₂ and 5% O₂.

(183) 3.4.2 Preparation of Cells for Transduction

(184) The day before seeding the cells, 12-well dishes were coated with Matrigel as described in reference NCP.003 (NCP.003: Culturing of iPS cells from CHM patient and controls). Before transduction of iPS cells with respective AAV2 or AAV8 viral vectors, the cells that are cultured on MEFs were seeded on Matrigel without MEFs (feeder free culturing). Cells were seeded at a density of 4.5+E5 to 6+E5 in 1 ml of iPS cell culture media in each well of a 12-well cell culture dish. Seeded cells were incubated at 37° C. in an environment supplied with 5% CO₂, 5% O₂.

(185) 3.4.3 Transduction

(186) To infect the iPS cells with viral vectors, cells were grown to approximately 50-60% confluence. (This can take 2-4 days in feeder free conditions). Once 50-60% confluence is achieved, one well of the 12-wells is dissociated and cell counts were performed to determine the total number of cells per well. Wells of the iPS cells were then infected with AAV vectors listed below at the predetermined MOI (see Table 18 and 19). Before transduction, the old iPS cell culture media from the plates was removed and a fresh 1 ml of iPS cell culture media was added in each well. Predetermined volumes of the virus from the stock were directly added to each well. See Table 18. and Table 19. For the information on total number of cells infected, MOI and the volume of virus used for infection. Cells were then incubated at 37° C. in an environment supplied with 5% CO₂, 5% O₂ for 18-24 hours. After 18-24 h of transduction, cells were observed under microscope for any abnormalities or cell death. At this point, another 1 ml of fresh iPS cell culture media was added to each well containing infected and uninfected cells and were further incubated for additional 18-24 hours at 37° C. in an environment supplied with 5% CO₂, 5% O₂. Cells were observed under the microscope before harvesting to evaluate any cell death or abnormal appearance.

(187) TABLE-US-00022 TABLE 18 Infection details and MOIs of next generation

AAV2.hCHMV2a, 2B, 3a, 3b Vectors and AAV2.hCHM.V1 vectors in CHM patient-derived iPS cells. Cell Viral stock Vector Vector Cell line Cell concentration used Used Line number density (vg/ml) (μL) MOI AAV2.V2a iPSC JB 588 6E5 2.16E+12 30 1E5 iPSC JB 588 6E5 2.16E+12 90 3E5 AAV2.V2b iPSC JB 588 6E5 7.4E+12 8.1 1E5 iPSC JB 588 6E5 7.4E+12 24.3 3E5 AAV2.V3a iPSC JB 588 6E5 4.82E+12 12.4 1E5 iPSC JB 588 6E5 4.82E+12 37.3 3E5 AAV2.V3b iPSC JB 588 6E5 5.91E+12 10.2 1E5 iPSC JB 588 6E5 5.91E+12 30.5 3E5 AAV2.V1 iPSC JB 588 6E5 4.47E+12 30.6 1E5 iPSC JB 588 6E5 4.47E+12 40.9 3E5

(188) TABLE-US-00023 TABLE 19 Infection dose of AAV8.V2a and AAV8.V1 vectors in three iPS cell lines derived from 3 different CHM patients. Cell Viral stock Vector Vector Cell line Cell concentration used Used Line number density (vg/ml) (μL) MOI Untreated iPSC JB 588 4.5E5 AAV8.V2a iPSC JB 588 4.5E5 1.04E+13 43 1E6 Untreated iPSC JB 500 4.5E5 AAV8.V1 iPSC JB 500 4.5E5 1.39E+13 32 1E6 Untreated iPSC JB 527 4.5E5 AAV8.V1 iPSC JB 527 4.5E5 1.39E+13 32 1E6 AAV8.V2a iPSC JB 527 4.5E5 1.04E+13 43 1E6

(189) Outcome measurement method-Western blot analysis was performed as described herein.

(190) Results

(191) 5.1 Expression of AAV2-hCHM V1, V2a, V2b, V3a, V3b in JB588 iPS cell line: Monoclonal human REP-1-specific antibody, detected one single ~75-80 kDa hREP-1 protein in the transduced JB 588 iPS cells (Data not shown). No band was observed in the case of the untreated control, confirming presence of the disease (data not shown). The intensity of REP-1 protein band at an MOI of 3E5 observed to be stronger in all vectors compared to an MOI of 1E5. Recombinant AAV2. hCHM viral mediated delivery of the hCHM gene, to iPS cells, resulted in a dose-dependent production of REP-1 protein. Probing of the blots with GAPDH antibody showed a band

of equal density in all lysates. GAPDH detected a protein at ~39 kDa. Both REP-1 and GAPDH antibodies detected only specific bands of expected molecular weight. No nonspecific bands were observed in the blots.

(192) Expression of AAV8-hCHM. V1, V2a in iPS cells: Monoclonal human REP-1-specific antibody, detected one single ~75-80 kDa REP-1 protein in the transduced JB527, JB500 and JB588 patient derived iPS cells (Data not shown). No protein band was observed in the case of the untreated control. (Data not shown). Probing of the blots with GAPDH antibody showed a band of equal density in all cell lysates including the cell lysates from untreated cells. Anti-GAPDH antibody detected a specific ~39 kDa protein band. Both REP-1 and GAPDH antibodies detected only specific protein bands at the expected size molecular weight. No detectable nonspecific protein bands were observed in the blot.

CONCLUSIONS

(193) The preliminary results presented in the current report revealed the following observations: Western blot analysis confirmed presence of CHM (lack of REP-1 protein) in each one of the three patient-derived iPSCs (JB588, JB500, JB527). In vitro expression studies demonstrated that infecting iPS cells from CHM patients with AAV2.hCHM. Version 2a, 2b, 3a, 3b and AAV2.hCHM Version1 (a current clinical trial candidate) induced the production of REP-1 protein at all tested MOIs. Infecting iPS cells with AAV8.hCHM. Version 2a and AAV8.hCHMVersion1 at an MOI of 1E6 resulted in production of REP1 protein in all three CHM iPS cell lines. Level of REP1 production was higher in the iPSCs infected with AAV8.hCHM.V2a than with AAV8.hCHM.V1. Example 7: Comparison of In Vivo Expression of AAV8.Codon Optimized.Human CHM Versus AAV.Native.Human CHM

(194) Gene therapy for a number of retinal diseases depends on efficient transduction of the appropriate target cells, which for choroideremia, are retinal pigment epithelium (RPE) cells and photoreceptor cells. This study report focuses on the comparison of in vivo expression induced by the native CHM sequence based construct, (Version 1) and four next generation transgene cassettes packaged into an AAV8 backbone in wild type mice. Here we evaluated AAV8 serotype for the purpose of improving gene transfer to photoreceptor cells.

(195) Our experiments were designed to answer the following questions: a. How would these vectors compare for in vivo transduction of photoreceptors: In particular, how efficiently would the next generation AAV8. CHM transduce photoreceptors after subretinal injection of the respective test article compared to version.1. b. Dose response: Would the next generation AAV8. CHM and AAV8. CHM-Version1 vectors differ in dose response of gene expression.

(196) Experimental Details:

(197) Plasmids and vectors were as described in Example 4. Mice (Animals): Wild type, CD1 mice were used to test the in vivo expression of CHM as assessed by production of REP-1 protein. CD1 mouse strain is an outbred Swiss mouse strain which colony we maintain in house. The details of the study are described under CAROT study protocol PCPR02.01.

(198) 3.3 Study Design (e.g. Treatment Groups)

(199) 3.3.1 Animal Husbandry: Both male and female mice (~3-4 months old) weighing ~20-30 gm were injected with the described test articles. Animals were housed in the University of Pennsylvania's John Morgan University Laboratory Animal Resources (ULAR) facility according to University of Pennsylvania's ULAR regulations. Mice were maintained on a 12-hour light/12-hour dark cycle. Food and water were provided ad libitum. All animals were identified with ear tag numbers.

(200) 3.4 Test material administration: The test article formulation provided by the CAROT Vector Core was used for dose administration. The test material was stored at -60 to -80° C. The test material was thawed on ice prior to dosing. For intra-ocular injections, the test article is diluted to the target concentration with phosphate-buffered saline as described in the formulation Table 20. A total of 60 µl of master mix was prepared.

(201) TABLE-US-00024 TABLE 20 Dose Formulation table for subretinal injections of test articles. Volume of In- Total Vector (μ) For PBS (μl) jected vol. Conc a total for a total Conc inj. Identification Lot # (Vg/ml) of 60 μl of 60 μl (vg) (μl) AAV8.V2a CT245 1.04E+13 1.92 58.1 5E8 1.5 μ3 19.2 40.8 5E9 1.5 μ3 AAV8.V2b CT244 1.11E+13 1.8 58.2 5E8 1.5 μ3 18 42 5E9 1.5 μ3 AAV8.V3a CT259 8.67E+12 2.31 57.7 5E8 1.5 μ3 23.1 36.9 5E9 1.5 μ3 AAV8.V3b CT255 1.36E+13 1.47 58.53 5E8 1.5 μ3 14.7 45.3 5E9 1.5 μ3 AAV8.V1 KA808 1.39E+13 1.44 58.6 5E8 1.5 μ3 14.4 45.6 5E9 1.5 μ3

Preparation of Injection Log Before Subretinal Injections:

(202) An injection log was maintained with the following information before subretinal injection of the test articles: Cage Number/mouse number Study Identification Strain Date of Birth Date of injection Name of investigator/injector Eye injected into (left or right) Injection material (vector/serotype) Dose and Volume Route of Administration (ROA)

(203) Subretinal injections: Injections were performed by Subretinal Injection by the Surgeon. In brief, animals were anaesthetized before injection. Subretinal injection of the test article was performed using Hamilton 33G syringe. The details of test articles and injections are described in Table 21. From the prepared injection master mix, a volume of 1.5 μl was administered, per injection. One eye per animal was injected with 5E8 vg/eye and the contralateral eye was injected with 5E9 vg/eye.

(204) TABLE-US-00025 TABLE 21 Subretinal injection scheme and injection doses Vector Dose Dose Volume Minimum (vg/eye) (μL/eye) Gr. Number of Test Material Right Left Right Left Dosing No. Animals ROA Identification Eye Eye Eye Eye Day (PD) 1 2 Subretinal AAV8.V2a 5+E8 5+E9 1.5 μl 1.5 μl Day 90- 120 2 2 Subretinal AAV8.V2b 5+E8 5+E9 1.5 μl 1.5 μl Day 90- 120 3 2 Subretinal AAV8.V3a 5+E8 5+E9 1.5 μl 1.5 μl Day 90- 120 4 2 Subretinal AAV8.V3b 5+E8 5+E9 1.5 μl 1.5 μl Day 90- 120 5 2 Subretinal AAV8.V1 5+E8 5+E9 1.5 μl 1.5 μl Day 90- 120 6 2 Uninjected Not Applicable N/A N/A N/A N/A N/A (N/A)

Outcome Measurement Methods

(205) Animal Sacrifice: a. After injecting the animals with the test articles, all animals were observed for 48 hours for any post injection related abnormalities. b. 21-35 days of post injection, the animals were observed for ocular abnormalities using ophthalmoscopy. c. 90-12-days post injection, the animals were sacrificed and eye tissues were collected for evaluating the production of exogenous REP-1 protein by SDS-PAGE followed by western blot analysis.

(206) Collection of Eye Tissue: Eye tissue for western blot analysis was collected after removing the lens from the eye using a sharp surgical blade. The eye (without the lens) was collected in freezer tubes that are labelled appropriately.

(207) Western Blot Analysis

(208) Briefly: 1. Preparation of Tissue Lysate

(209) a. Ocular tissue of animals injected with 2 different doses of next generation AAV8. CHM and AAV8.V1 along with the uninjected control animal tissues, were collected after 21-35 days of injection by sacrificing the animals. b. Tissues were then lysed on ice using RIPA buffer with protease inhibitors.

(210) c. Tissue lysates were cleared by centrifuging at 13,000 rpm for 10 min.

(211) 2. Quantification and Preparation of Proteins

(212) a. Protein quantification of the cell lysates was carried out using ThermoFisher Micro BCA™ Protein Assay Kit following manufacturer's instructions. b. Protein concentration was determined by taking OD reading at 562 nm. c. To evaluate the in vivo expression of CHM, between 20-40 ug of measured protein was loaded on 4-12% Bis-Tris gels.

(213) 3. SDS-PAGE and Western Blot

(214) The protein gels were transferred on to a nitrocellulose membrane, blocked in milk and incubated with the primary antibodies. Anti-human REP-1 2F1 antibody (2F1, 1:1000 dilution) and/or anti-GAPDH antibody (1:1000 dilution); were used as primary antibodies. After washing the

blot, HRP conjugated anti-mouse IgG antibody and/or anti-rabbit IgG antibody at a concentration of 1:5000 were used as secondary antibodies. The blots were developed by chemiluminescence using ECL reagents according to the manufacturer's instructions.

(215) 4. Controls

(216) a) Loading controls: Anti-GAPDH antibody was used as loading controls to demonstrate equal loading of protein in each well of the gels. Anti-GAPDH antibody detects a protein of ~39 kDa. b) Positive control: AAV2. V2a transduced COS-7 cell lysates were used as positive controls. c) Negative control: Ocular tissues of uninjected animals were used as negative control.

(217) Sample Value Determination

(218) Quantification of Western blot analysis using Image J software. In brief, densitometric evaluations presented in this report are normalized first, to the levels of endogenous expression of GAPDH protein of the corresponding sample. Then the expression levels are normalized again, to the average REP-1 expression level of uninjected control.

(219) The details of densitometric evaluations and fold change calculations to represent the expression of REP-1 protein are presented as Table 22 and 23.

(220) The description in brief:

(221) 1. In table 22 and 23, Column 2 shows the raw values of REP-1 protein and column 3 shows the raw values of GAPDH protein.

(222) 2. GAPDH value of each samples was first normalized to the GAPDH values of animal-1 of AAV8.V1 and are shown in Table 22 in 4th column.

(223) 3. The values of each sample were also normalized to the GAPDH values of animal-2 of AAV8.V1 and are shown in Table 22 in 5th column.

(224) 4. The REP-1 values (Column 2) are then normalized to either to the GAPDH normalized to animal 1 (column 4) or to the GAPDH previously normalized to animal 2 (column 5). These are represented in column 6 and 7 respectively.

(225) 5. The normalized REP-1 values are then converted to fold change.

(226) 6. The respective REP-1 values are normalized to expression of REP-1 either in animal 1 or animal 2 of the AAV8.V1 injected group and are expressed as fold change (column 8 and 9)

(227) 7. Column 10 represents the average fold change in REP-1 protein expression.

(228) Results

(229) Comparison of the CHM expression using the native CHM AAV8.V1 versus the codon optimized CHM vectors: AAV8. V2a, V2b, V3a and V3b. Wild type CD1 mice were injected with two different doses of the each AAV8 vector: a high dose of 5E9 vg/eye and a low dose of 5E8 vg/eye. Following results describe the levels of REP1 protein after injection with high and low doses of AAV8.V1, AAV8.V2a and AAV8.V3a.

(230) Comparison of the expression of AAV8. V1 versus AAV8. V2a and AAV8.V3a (vectors with stuffer) in animals injected with high dose (5E9 vg/eye) of viral vector. Western blot analysis with human anti REP-1 antibody detected a ~75-80 kDa hREP-1 protein band in both (low and high dose injected) ocular tissues of each animal treated with either the next generation AAV8. V2a or V3a or the original AAV8. Version1. A very faint (minimal) band is observed in the case of the uninjected control mice, both. A band of increased intensity was observed in tissues that were transduced with next generation vectors (AAV8V.2a and AAV8.V3a) compared to the tissues transduced with Version.1. Anti-GAPDH antibodies showed a ~39 kDa band of equal density in all lanes of the western blot including the uninjected controls. Pre-stained protein marker is used to compare the molecular weights of protein of interest. Densitometric evaluation (quantification of the expression level) of the blots using ImageJ software demonstrated that production of REP-1 was increased in animals injected with one of the next generation AAV8. high and low doses constructs (V2a or V3a). (See Table 22 for values.)

(231) TABLE-US-00026 TABLE 22 Quantified REP-1 protein production results for treatment with high dose (5E9vg) AAV8 V1, V2a and V2b REP1 REP1 GAPDH GAPDH Normalized

Normalized Fold Fold Average Normalized Normalized to respective to respective change change in fold to GAPDH to GAPDH GAPDH GAPDH in REP-1 REP-1 change of Version 1 of Version 2 (normalized (normalized (normalized (normalized in REP-1 GAPDH REP-1 Animal 1 Animal 2 to animal 1) to animal 2) to animal 1) to animal 2) expression AAV8.V2a 12758.589 10058.359 1.055 1.190 9533.571 8450.521 2.335 3.742 3.038 AAV8.V2a 11885.518 13247.510 0.982 1.108 13489.242 11956.812 3.303 5.294 4.299 AAV8.V3a 12139.418 15542.551 1.003 1.132 15495.152 13734.843 3.794 6.081 4.938 AAV8.V3a 11113.640 7274.388 0.918 1.036 7921.575 7021.653 1.940 3.109 2.524 AAV8.V1 12102.397 4083.761 1.000 1.128 4083.761 3619.830 1.000 AAV8.V1 10727.518 2258.477 0.886 1.000 2547.932 2258.477 1.000 REP-1 expression values untreated animal was negligible see the values below: Name REP-1 GAPDH Uninjected Animal-1 651.678 16633.539 Animal-2 253.778 13025.397

(232) TABLE-US-00027 TABLE 23 Quantified REP-1 protein production results for treatment with low dose (5E8 vg) AAV8 V1, V2a and V2b REP1 REP1 GAPDH GAPDH Normalized Normalized Fold Fold Average Normalized Normalized to respective to respective change change in fold to GAPDH to GAPDH GAPDH GAPDH in REP-1 REP-1 change of Version 1 of Version 2 (normalized (normalized (normalized (normalized in REP-1 GAPDH REP-1 Animal 1 Animal 2 to animal 1) to animal 2) to animal 1) to animal 2) expression AAV8.V2a 11815.489 11194.037 0.809 0.934 13844.203 11984.286 11.123 29.133 20.128 AAV8.V2a 12889.418 7162.924 0.882 1.019 8120.634 7029.658 6.524 17.089 11.806 AAV8.V3a 13088.418 1516.506 0.896 1.035 1693.128 1465.663 1.360 3.563 2.462 AAV8.V3a 9201.075 593.192 0.630 0.727 942.084 815.519 0.757 1.982 1.370 AAV8.V1 14612.782 1244.678 1.000 1.155 1244.678 1077.460 1.000 AAV8.V1 12649.610 411.364 0.866 1.000 475.206 411.364 1.000 REP-1 expression values untreated animal was negligible see the values below: Name REP-1 GAPDH Uninjected Animal-1 694.263 15930.368 Animal-2 254.364 13896.246

(233) Comparison of the expression of AAV8.V1 versus AAV8.V2a and AAV8.V3a in animals injected with low dose (5E8 vg/eye) of viral vector

(234) Human anti REP-1 antibody, the Western blot analysis of the ocular tissues of animals injected with next generation AAV8.V2a, V3a and AAV8. Version1 at a dose of 5E8 detected a ~75-80 kDa hREP-1 protein band in tissues of injected mice. A faint (minimal) band of REP-1 was observed in ocular tissue lysates of the uninjected control mice, both. A band of increased intensity was observed in tissue lysates that are transduced with next generation vectors compared to the lysates that are transduced with Version1. Anti-GAPDH antibody detected an equal intensity protein band at ~39 kDa in all cell lysates. This data demonstrates that delivery of next generation V2a CHM through AAV8 results in robust levels of REP-1 protein in comparison with levels produced after injection of AAV8.V3a or AAV8.V1.

(235) Densitometric evaluation (quantification of the expression level) of the blots using ImageJ software further demonstrate an increased production of REP-1 in animals injected with next generation AAV8.CHM constructs (especially V2a) compared with Version 1. See Table 23 for values.

(236) Expression of AAV8. V2b in CD1 Mice

(237) This current study and the evaluation of lambda stuffer's effect on AAV vector production by qPCR titer analysis were carried out simultaneously. We performed all the animal injections for the in vivo expression study as described in the study protocol PCPR.02 and all samples were harvested. After the qPCR study on the lambda stuffer element was concluded (described above), we decided to carry out the Western blot experiments only to test the expression of AAV vectors without the stuffer such as AAV8.2b and AAV8.3b and exclude them from further analysis (such as comparison with Version 1).

(238) Human anti-REP-1 antibody detected a protein of ~75-80 kDa in ocular tissues of CD-1 mice injected with AAV8.2b at 5E9 (High dose) vector genome copies (FIG. 12A). Animals injected with AAV8.2b at 5E8 (Low dose) showed a very faint protein band at ~75-80 kDa (FIG. 12A).

Lysates of ocular tissues from uninjected control animals did not show the presence of REP-1 protein. Anti-GAPDH antibody detected a protein of ~39 kDa in all ocular tissue lysates including the uninjected controls. This data may establish the minimal dose for AAV8.2b.

(239) Expression of AAV8.V3b in CD1 Mice

(240) We performed a Western blot analysis on ocular tissues of AAV8.3b injected CD1 mice (2 mice/group) with anti-REP-1 antibody, which revealed the presence of a protein of ~75-80 kDa in one eye injected with low dose and in both eyes injected with high dose of AAV8.3b. In the ocular tissues of uninjected mice there was no REP-1 expression detected (FIG. 12B). The level of REP-1 produced was dose dependent in animals injected with AAV8.3b. Injection with high dose of AAV8.3b (5E9 vector genomes) induced a higher amount of REP-1 compared with the low dose injected eyes (5E8 vector genomes). Anti-GAPDH antibody detected a protein of ~39 kDa in ocular tissue lysates of all injected and uninjected animals.

(241) These results revealed the following observations:

(242) 1) The next generation vectors AAV8. Version2a, 2b, 3a and 3b are able to transduce ocular tissues efficiently. 2) Expression of the transgene (codon optimized CHM) was detectable for all of the next generation vectors. 3) Expression of transgene (codon optimized CHM) is dose dependent. 4) AAV8. Version2a and AAV8. Version.2b induced an increased production of REP-1 protein compared to AAV8. Version 1 in ocular tissues of CD-1 mice. 5) There is variation in the exact level of production of the transgenic protein between eyes injected with the same dose reflecting the variability in the surgical delivery procedure. However, differences in levels are large between the low (5E8) and high (5E9) doses. 6) AAV8.CHM.V2a and AAV8. V3a result in much higher levels of REP-1 protein production than AAV8.V1 after in vivo administration of high dose (5E9 vg) vector subretinally in mice.

(243) All publications cited in this specification are incorporated herein by reference. Similarly, the SEQ ID NOs which are referenced herein and which appear in the appended Sequence Listing are incorporated by reference. While the invention has been described with reference to particular embodiments, it will be appreciated that modifications can be made without departing from the spirit of the invention. Such modifications are intended to fall within the scope of the appended claims.

(244) TABLE-US-00028 SEQ ID NO (SID) Sequence 1 codon optimized REP-1 2 AA from SID 1 3 native REP-1 4 AA from SID 3 5 codon optimized REP-1 with modifications 6 AA from SID 5 7 p584 8 p584-REP-1 optimized 9 codon optimized CNGA3 10 AA from SID 9 11 codon optimized CNGA variant X1 12 AA from SID 11 13 native CNGA3 14 AA from SID 13 15 native CNGA3 variant X1 16 codon optimized CNGA3 with modifications 17 codon optimized CNGA3 variant with modifications 18 native CNGA3 with modifications 19 native CNGB3 20 AA from SID 19 21 CNGB3 with modified ORF 22 AA from SID21 23 CNGB3 with modified ORF with modified ends 24 AA from SID 23 25 Plasmid Version 2a 26 Plasmid version 2b 27 Plasmid version3a 28 Plasmid version 3b 29 Plasmid version 1

Claims

1. A method for treating achromatopsia in a human subject in need thereof, said method comprising administering to the human subject subretinally an AAV vector comprising an AAV capsid and a nucleic acid sequence packaged therein, wherein the nucleic acid sequence comprises i) AAV ITR sequences, ii) a nucleic acid sequence encoding human cyclic nucleotide gated channel alpha 3 (CNGA3), and iii) expression control sequences that direct expression of the CNGA3 in a host cell, wherein the nucleic acid sequence encoding the human CNGA3 sequence comprises SEQ ID NO: 9 or SEQ ID NO: 11.

2. The method of claim 1, wherein nucleic acid sequence encoding human CNGA3 sequence comprises SEQ ID NO: 9 or SEQ ID NO: 11.

3. The method of claim 1, wherein the expression control sequences comprise an ocular cell-specific promoter.
 4. The method of claim 1, wherein the AAV vector is administered in combination with another therapy.
 5. The method of claim 1, wherein the AAV vector is administered at a dosage of about 10⁹ to about 10¹³ vector genomes (VG).
 6. The method of claim 1, wherein the AAV vector is administered in a volume of about 100 μ L to about 500 μ L.
 7. The method of claim 1, wherein the AAV vector is administered more than once.
 8. The method of claim 1, wherein the expression control sequences comprise a chicken β -actin (CBA) promoter with cytomegalovirus (CMV) enhancer elements.
 9. The method of claim 1, wherein the expression control sequences comprise a rhodopsin kinase promoter.
 10. The method of claim 1, wherein the AAV capsid is an AAV2 capsid, an AAV5 capsid, an AAV8 capsid, an AAV9 capsid, an AAV8 bp capsid, an AAV7m8 capsid, or a variant thereof.
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